



Aspect-Based Sentiment Analysis of Code-Switched Consumer Reviews for Local Coffee and Matcha Businesses in Davao City

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Chapter 1

Introduction

1.1 Background of the Study

Cafe culture, particularly matcha and coffee shop hopping, became increasingly popular in the Philippines. This trend led to a significant rise in demand for Matcha and Coffee, prompting multiple shops to open in the city. In urban areas such as Davao City, this rapid increase has led to a highly competitive and saturated market [18]. Consequently, consumers are given a hard time efficiently comparing cafés. This concern becomes even more significant as people from different parts of the country travel to Davao City specifically to explore its growing coffee and matcha scene.

Despite the availability of online reviews on platforms such as Google Maps, Facebook, and Instagram, these reviews are often unstructured, not normalized, fragmented, and overwhelming in volume. The presence of many positive reviews and high rankings for a business or its products may make it more likely that a potential consumer will feel confident about making a purchase [25]. However, the large volume of reviews requires consumers to manually sift through them, making decision-making time-consuming and inefficient. Furthermore, some of these reviews may contain biased, misleading, or even irrelevant information, which can negatively affect consumer choices and business reputations [27].

This major limitation stems from the inability of existing systems to effectively organize, analyze, and interpret large volumes of unstructured review data. Although thousands of reviews are available across platforms, consumers must manually read them in raw, unstructured text, which is both time-consuming and inefficient. Moreover, this issue is further complicated by code-switching, informal language, and inconsistent expressions, making it difficult, and in some cases impossible, for traditional analysis methods to accurately capture sentiment and meaning [32][28].

To address these challenges, prior studies have used automated data collection methods, such as Application Programming Interfaces (APIs), to convert raw reviews into structured datasets for analysis, thereby improving scalability and data reliability [34]. Sentiment analysis, on the other hand, relied on lexicon-based techniques such as

VADER, most especially in the earlier years [8]. Although it was useful in English-written texts, the techniques were found to be limited in handling multilingual and code-switched texts commonly found in Philippine contexts. As a result, more advanced learning-based models, particularly multilingual transformer models such as XLM-RoBERTa, were introduced to better interpret complex language patterns and improve classification performance [32]. Multiple studies have also utilized aspect-level analysis through Aspect-based Sentiment Analysis (ABSA) to extract more detailed insights by evaluating specific attributes [46]. Extensive manual annotation was also greatly reduced by using hybrid strategies, including Active learning, which has been found to improve performance [48]. The aforementioned approaches offer several advantages for addressing the limitations of traditional text and review analysis, particularly when integrated into a unified process. While standard Application Programming Interfaces (APIs) offer structured data, they often impose severe extraction limits. Thus, customizing an automated web scraping process ensures greater scalability and sufficient data volume when handling large datasets.

Furthermore, the application of Aspect-Based Sentiment Analysis (ABSA) enables the decomposition of reviews into distinct attributes, yielding more precise insights. To improve efficiency and reduce reliance on large labeled datasets, techniques such as Active Learning prioritize the most informative samples for annotation, significantly reducing labeling effort while maintaining performance [37]. Lastly, data augmentation strategies, such as regulated synthetic data generation, help address issues of data scarcity and class imbalance [20]. By using these approaches, a framework is created that transforms unstructured, multilingual, and context-rich texts into more meaningful insights.

Though the Davao City cafe and matcha shop scene is rapidly expanding, there is a distinct lack of research utilizing advanced Natural Language Processing (NLP) techniques to evaluate consumer preferences. The study fills the gap by implementing XLM-RoBERTa a Aspect-Based Sentiment Analysis (ABSA) framework, enhanced by Active Learning, specifically for Taglish and Bislish reviews.

1.2 Problem Statement

This study investigates the challenge of effectively analyzing unstructured, multilingual, and context-rich consumer reviews, specifically those that contain code-switching and informal language. Existing platforms generate large volumes of raw, unstructured textual data that is challenging to interpret using traditional methods, hindering the extraction of meaningful, reliable results. Specifically, this study aims to answer the following questions:

1. What aspect categories characterize the multilingual consumer reviews of local coffee and matcha businesses in Davao City, and how does the code-switched nature of the text affect aspect-level sentiment classification?
2. How does the proposed Multilingual Transformation model perform compared to lexicon-based approaches on code-switched and informal review texts?
3. What are the aspect categories that emerged from the consumer reviews of Davao City and matcha establishments, and what are the prevailing consumer sentiments associated with these discovered dimensions?

4. How can active learning and synthetic data generation improve model performance and reduce annotation effort?
5. How can the proposed model be evaluated and integrated into a system that supports consumer decision-making and business analysis?

1.3 Objectives

The main objective of this study is to develop an XLM-RoBERTa-based system for aspect-based sentiment analysis that efficiently handles multilingual and code-switched consumer reviews. The study also wishes to accomplish the following objectives:

1. To identify the aspect categories present in multilingual consumer reviews of local coffee and matcha businesses in Davao City and examine how code-switching affects aspect-level sentiment classification.
2. To compare the performance of the proposed XLM-RoBERTa-based model against lexicon-based approaches on code-switched and informal review texts.
3. To identify the aspect categories that emerged from consumer reviews of Davao City coffee and matcha establishments and determine the prevailing consumer sentiments associated with each discovered dimension.
4. To implement active learning strategies and synthetic data generation to reduce annotation effort while maintaining classification accuracy.
5. To evaluate the proposed model and integrate it into a system that supports consumer decision-making and business analysis.

1.4 Significance

This study presents significant contributions to various stakeholders by addressing the challenge of analyzing large volumes of unstructured, multilingual, and context-rich consumer reviews.

1. **For Consumers:** The rapid growth of cafe culture in Davao City has led to a highly competitive and saturated market [18]. Consequently, consumers face difficulties in efficiently comparing different establishments. An Aspect-Based Sentiment Analysis (ABSA) system alleviates the burden of manually processing thousands of reviews. Rather than reading extensive texts, consumers are provided with concise, data-driven summaries of specific aspects they find most valuable.
2. **For Local Coffee and Matcha Businesses:** Aggregated star ratings often fail to provide actionable insights. For instance, a single review may contain positive sentiments regarding a product's taste but negative sentiments toward the establishment's customer service or ambiance. Decomposing reviews into specific features enables businesses to accurately pinpoint operational strengths and identify specific areas requiring improvement.

3. **For Academic Research:** This study fine-tunes XLM-RoBERTa on Taglish and Bislish, code-switched dialects prevalent in the Philippines. Data scarcity constrains model performance; this work employs two complementary techniques to mitigate it. Active Learning identifies the most informative unlabeled examples for manual annotation based on model uncertainty [48]. Synthetic data generation then augments the training set, reducing the annotation burden while maintaining dataset quality.

1.5 Scope and Limitations

This study analyzes consumer reviews of local coffee and matcha businesses in Davao City. To ensure a comprehensive analysis, data collection encompasses all publicly available reviews for the selected establishments from their inception up to the date of extraction, with no minimum threshold or maximum limit imposed on the dataset volume. Due to the extraction limits of Google Places API, reviews are acquired through custom HTML scraping to ensure sufficient data volume and will be parsed into JSON format. The dataset consists of multilingual, code-switched text spanning English, Tagalog, and Bisaya. A fine-tuned XLM-RoBERTa model performs aspect-based sentiment classification across predefined dimensions. Active learning is applied to identify uncertain samples for manual annotation, reducing labeling effort, while LLM-based synthetic data generation addresses class imbalance in underrepresented aspect-sentiment pairs.

Model collapse and potential loss of linguistic diversity resulting from synthetic data generation are not addressed in this study. Synthetic data are used only to supplement underrepresented classes, and human-annotated reviews remain the primary training source. Data collection is constrained by the inherent limitations of web scraping, such as dynamic website structure changes (DOM updates) and potential IP blocking. The finalized model will be integrated into a system. However, the underlying machine learning pipeline focuses exclusively on textual data. Image and video content in reviews are not processed.

1.6 Definition of Terms

1. **Active Learning (AL):** A machine learning framework that reduces manual annotation effort by selecting only the most informative or uncertain samples for labeling, rather than processing an entire dataset.
2. **Application Programming Interface (API):** A set of protocols that automates the extraction of large volumes of online consumer reviews into structured formats such as JSON.
3. **Aspect-Based Sentiment Analysis (ABSA):** A natural language processing technique that identifies specific attributes of a subject in text, such as taste, price, service, and ambiance, and determines the sentiment expressed toward each.
4. **Code-Switching:** The practice of alternating between two or more languages

within a single text, as seen in “Taglish” (Tagalog-English) and “Bislish” (Bisaya-English), the dominant mixed-language variants in the dataset.

5. **Lexicon-Based Approaches:** Sentiment analysis methods that classify text using predefined word lists where each entry carries a positive or negative polarity score, with VADER as a representative example.
6. **Model Collapse:** A sustained decline in model performance caused by training predominantly on synthetic rather than human-generated text, resulting in reduced linguistic diversity over successive iterations.
7. **Multilingual BERT (XLM-RoBERTa):** A pre-trained transformer model that processes text across multiple languages in context, capable of interpreting code-mixed reviews written in Filipino dialects and English.
8. **Synthetic Data Generation:** The process of producing artificial training examples from a small seed set of real reviews using large language models, applied to address data scarcity and class imbalance.
9. **Unstructured Data:** Raw text lacking a predefined schema or organization, such as user-generated online reviews, which resists systematic processing without specialized tools.
10. **Web Scraping:** An automated technique that extracts data from websites by parsing HTML to gather raw online reviews at scale.

Chapter 2

Review of Related Literature

2.1 Local Coffee and Matcha Market

The rapid increase of cafe culture in the Philippines has significantly transformed local beverage shops [18]. Urban cities like Davao City have experienced exponential growth in coffee and matcha, creating a highly saturated, competitive environment. Local consumers, particularly the Gen Z demographic, are a key driver of this market.

1. *Market Competition:* The local coffee shop industry in Davao City has risen, evolving into a competitive service environment. When new establishments open, the market is quick to absorb them. However, sustaining the business requires distinct differentiations. Local cafe owners are leveraging locally sourced products (Mt. Apo coffee and award-winning Davao cacao) to establish their own unique market. This stimulates demand for both local and national brands and local competitors [3].
2. *Influencers:* Consumer decisions in Davao City are dependent on digital social proof [26]. Local consumers usually conduct online research before making purchasing decisions. Hence, consumer choices are heavily influenced by online reviews, social media recommendations, and peer opinions, making reputation management and digital influence necessary for market success in the region.
3. *Digital Engagement:* Gabato et al. [18] investigated the preferred digital marketing approaches among Gen Z college students in Davao City. They approached the problem by understanding how different digital attributes drive consumer interaction. Using a conjoint analysis approach through the PAPRIKA method, their study assessed attributes including social media, publicity types, and content types. They demonstrated that in the specific context of Davao City's Gen Z student market, content type (26.7%) and platform choice (24.7%) are the most influential factors. This proves that this demographic favors engaging, interactive content over traditional online advertisements [18]
4. *Consumer Decision-Making:* Gen Z consumers in Davao City often consider their experience and the physical ambiance of a coffee shop before they return. According to a study by Alcraz et al. [4], they modeled consumer behavior using the "Theory

of Planned Behavior” to assess Gen Z tourists’ attitudes toward local coffee shops. They found a correlation between experiences and visit intention (0.693) and a correlation of (0.614) between attitudes and visit intention [4].

2.2 Text Analysis of Consumer Reviews

Consumer reviews have become a primary source of public sentiment in digital markets, and processing them effectively requires both categorizing the type of feedback being expressed and understanding its structure [30] [21]. Lehto et al. [30] addressed this by analyzing unstructured consumer feedback from virtual travel agencies and categorizing reviews into three categories (complaints, compliments, and satisfaction). Classifying text along these polarities enables businesses to isolate specific problem areas and convert large volumes of feedback into actionable insights for managing consumer loyalty.

Structuring the text itself requires further decomposition. Herhausen et al. [21] noted that business researchers often lack coherent frameworks for processing digital stakeholder communication, and proposed applying a message design logic lens to break reviews into three components (background, situation, and intention). Organizing unstructured language into these elements provides researchers with a replicable method for translating consumer-generated text into measurable insights [21].

However, the explosion of user-generated content (UGC) on platforms like Google Maps has turned local market analysis into a massive ”Big Data” challenge. While consumer reviews provide valuable feedback, the huge volume of unstructured text renders manual analysis employing traditional qualitative methods impractical [52]. Almuayqil et al. (2022) [5] confronted the challenge of extracting reliable sentiment from large bodies of redundant and inconsistent review data. Pairing sentiment analysis algorithms with a preprocessing pipeline that included tokenization, lowercasing, and punctuation removal, they found that rigorous data normalization substantially improved a classifier’s ability to handle noisy text and detect sentiment polarity accurately [5].

Extracting meaningful insights from thousands of Davao City coffee and matcha shop reviews cannot rely solely on manual reading [27]. Natural language at scale also introduces problems of sparsity, dimensionality, and ambiguity that rule-based models and simple frequency counts are ill-equipped to handle [2]. Automated preprocessing pipelines paired with advanced sentiment analysis are, therefore, necessary to produce reliable, localized findings from this volume of consumer-generated [40] [32].

2.3 Code-Switching

Code-switching presents consistent uses in examining reviews from the Philippines. This refers to how writers or speakers switch between English, Tagalog, and other regional languages within the same text. [38] [14]. Mixed-language varieties such as Taglish (Tagalog-English) and Bislish (Bisaya-English) present particular difficulties for NLP models trained on monolingual English corpora [22]. A Davao-based review, for instance, might use the Bisaya term *lami* (meaning ”delicious” [55]) alongside English or Taga-

log commentary about service. A standard English lexicon would register no positive sentiment in such a sentence at all [38].

There are other issues where code-switching involves the simultaneous blending of distinct grammatical rules and syntactic structures, which standard computational models are not designed to handle [38] [54]. Patil et al. [40] noted that most sentiment analysis tools in big data contexts are built around monolingual English, leaving them poorly suited for low-resource languages, particularly when multiple languages appear within a single post. Conventional feature extraction methods tend to treat hybridized or region-specific expressions as out-of-vocabulary errors, discarding the semantic content that gives those expressions their meaning.

Other works on Philippine consumer data confirm this. Cosme and De Leon [14] discovered that lexicon-based tools perform poorly on Filipino-English reviews because single-language systems cannot reliably determine polarity when inputs shift between languages mid-sentence. Ang Ngo Ching et al. [8] reached a similar conclusion. Observing that English-only dictionaries lack the syntactic coverage necessary to process Taglish text without significant accuracy loss.

Through these limitations, techniques that can tolerate inconsistencies in grammar, vocabulary, and context across languages are essential. Maceda et al. [32] make this point in the context of code-mixed social data, where the input rarely conforms to the clean, monolingual text that traditional models assume. Multilingual transformers are suited to this environment, as their shared subword vocabularies and self-attention mechanisms enable fine-tuning on localized, code-mixed corpora, enabling sentiment classification across blended language boundaries [54] [32].

2.4 Lexicon-Based Approaches

Sentiment analysis has long relied on tools such as VADER, developed by Hutto and Gilbert in 2014 [24]. The model combines a curated lexicon with grammatical rules for negation and degree modifiers, and it was designed specifically for the messiness of social media: slang, emojis, and punctuation stacking. On standard English data, it performs well, achieving an F1-score of 0.96 [24]. That performance held in practice as Mathayomchan and Taecharungroj (2020) [34] used VADER to analyze 935,000 Google Maps reviews from 5,010 UK restaurants. Mapping lexical signals across food, service, atmosphere, and value, they found that food quality is the main driver of five-star ratings, and poor service accounts for most one and two-star reviews, which is clean and useful for restaurant owners.

The limitations of this approach become apparent, however, when the input deviates from standard English. Ang Ngo Ching et al. [8] examined this problem in the context of Philippine online reviews, where Filipino-English code-switching and code-mixing are pervasive features of written discourse rather than exceptional occurrences. The researchers evaluated six sentiment analysis tools, including VADER, against an annotated dataset of 1,000 reviews sourced from Shopee and Google Maps, assessing performance at both the sentence and review levels. VADER recorded a macro-F1 score of 0.30, repre-

sentencing a 35% decline from its performance on standard English text. This is due to the model’s English-only lexicon, which lacks the vocabulary and syntactic coverage necessary to process Taglish, the hybrid register in which Filipino speakers routinely alternate between Filipino and English [8]. The findings indicate that lexicon-based methods are insufficient as primary classification tools for multilingual, code-switched text in localized contexts such as the Philippines [8].

Furthermore, when compared to learning-based models, lexicon methods are particularly ineffective for highly localized data, such as consumer reviews in Davao City. Standard English lexicons completely miss positive sentiment when encountering unique regional syntax and Bisaya terms, such as *lami* (Bisaya adjective meaning delicious [55]). Ortega [38] addressed this linguistic barrier by developing a neural network-based sentiment analyzer trained specifically on Bisaya-language text. The study demonstrated that both Bisaya and mixed Tagalog-English syntax consistently avoid standard dictionary mappings, leading to severe misclassification of consumer opinions when processed with conventional lexicon-based tools [38]. These findings reinforce a conclusion consistent with prior literature. While lexicon-based algorithms such as VADER offer a useful baseline, their dependence on monolingual English dictionaries makes them inadequate for text characterized by regional code-switching. This limitation, specifically the failure of English lexicons to account for Bisaya and Taglish syntax, is the reason this study favors learning-based models over traditional approaches.

2.5 Learning-Based Approaches

2.5.1 Machine Learning

Where lexicon-based approaches fail against ambiguity and context dependency, machine learning methods offer greater adaptability. Pratmanto et al. [43] applied Naive Bayes classification to Indonesian e-commerce reviews scraped from the Google Play Store, focusing on user feedback for the Shopee platform. Before classification, the text data was preprocessed through tokenization, stop-word removal, and stemming, then partitioned into training and testing subsets. The model achieved 96.67% accuracy, indicating that probabilistic classifiers can handle large volumes of user-generated reviews without the need for manually constructed sentiment lexicons [43].

Pakpahan et al. [39] addressed the challenge of granular, aspect-level analysis by evaluating four machine learning algorithms, which are Naive Bayes, SVM, Random Forest, and K-Nearest Neighbors, on 20,000 Indonesian reviews from Shopee and Tokopedia. Word2Vec was used for feature extraction to capture semantic relationships between terms. Of the four models, SVM performed best, reaching 91% accuracy in aspect extraction and 90% in sentiment polarity classification [50].

Ulhaq and Suprayogi [50] took a similar comparative approach, applying SVM, Random Forest, and Multilayer Perceptron (MLP) architectures to over 10,000 preprocessed Tokopedia reviews vectorized using TF-IDF. The MLP model achieved the highest accuracy of 87.21%, narrowly surpassing SVM, indicating that neural classifiers offer a modest but consistent advantage over traditional methods on preprocessed consumer text

[50].

However, while traditional machine learning algorithms demonstrate high performance on standardized datasets, their reliance on static feature extraction methods (such as TF-IDF) limits their ability to capture deep sequential context and complex semantic relationships [40]. Since these traditional methods treat text as isolated tokens or static vectors, they lack the representational capacity required to process severe linguistic complexity and code-switching effectively [54]. Consequently, these traditional architectures are insufficient for handling the highly nuanced, code-switched syntax (Taglish and Bislish) present in localized Davao City reviews. This critical limitation in processing dynamic, mixed-language structures naturally necessitates the deployment of advanced learning-based approaches, such as Multilingual Transformers, which are explicitly designed to capture these complex contextual relationships [32].

2.5.2 Multilingual Transformers

Where lexicon-based and traditional machine learning methods fall short in handling linguistic complexity, the literature points toward deep learning as the more capable alternative. Multilingual transformer models, particularly mBERT and XLM-RoBERTa, have gained considerable traction in code-switched text analysis. Their effectiveness stems from transfer learning in both models, which are pre-trained on large multilingual corpora [54] [40], which allows them to recognize sentiment across mixed-language patterns such as Taglish and Bislish before being adapted to a specific classification task [32].

The core difficulty in processing code-switched text is that most NLP models are trained on monolingual data, leaving them poorly equipped for sentences that blend two or more languages within a single utterance. Winata et al. [54] examined this problem by evaluating mBERT and XLM-RoBERTa across several code-mixed datasets. Their findings confirmed that while both models carry strong cross-lingual capabilities through shared subword vocabularies and self-attention mechanisms, fine-tuning on task-specific, code-switched data produces substantial gains in classification performance, establishing these architectures as well-suited backbones for mixed-language tasks [54].

The necessity of localized fine-tuning was directly taken up in the Philippine context by Maceda et al. [32], who analyzed public sentiment on higher education policy using social media data rich in Filipino-English code-mixing. They adapted mBERT's pre-trained linguistic knowledge to the syntactic patterns of their Taglish corpus, ultimately achieving an F1-score of 81.14% and outperforming Support Vector Machine baselines by a considerable margin [32]. The result reinforced the practical value of domain-specific fine-tuning when the target language environment departs substantially from standard monolingual text.

Furthermore, the sheer scale and variability of social media data present obstacles. Patil et al. [40] assessed sentiment analysis frameworks against unstructured data from social media platforms, where textual noise, informal jargon, and inconsistent grammar routinely disrupt conventional feature extraction. Their comparative evaluation found that transformer-based models consistently outperformed traditional machine learning methods in accuracy and robustness across datasets with high linguistic variability [40].

Salve and Tubil [45] demonstrated that XLM-RoBERTa performs well on Southeast Asian language contexts, specifically in handling the kind of rapid language switching and non-standard syntax that appears in bilingual, low-resource settings. In the Philippines, consumer-generated text mixes Tagalog and English in which creates problems for most natural language processing models. Informal spellings, slang, and unstable sentence structures produce ambiguities that some models struggle to resolve. To address this, Salve and Tubil [45] fine-tuned XLM-RoBERTa Large on 10,000 Taglish reviews, using a custom SentencePiece tokenizer and explicit ambiguity tagging at both the token and sentence levels. The model outperformed both its Base variant and mBERT, reaching a testing precision of 97.62%, an F1-score of 95.77%, and an ROC-AUC of 97.50%. These results suggest that XLM-RoBERTa’s pre-training across more than 100 languages gives it a meaningful advantage when classifying mixed-language text, which is directly relevant to the code-switched review data collected for this study [45].

These considerations bear directly on the analytical demands of this study. Consumer reviews from Davao City coffee and matcha shops are written in Bisaya-English and Tagalog-English [3][26][18]. These varieties use region-specific vocabulary, such as *lami* (a Bisaya adjective meaning ”delicious” [55]), that standard lexicon-based tools are unable to interpret. Thus, while this research is theoretically grounded in the Aspect-Based Sentiment Analysis (ABSA) framework utilized by Maceda et al. [32]. The study will implement XLM-RoBERTa as the primary classification engine. By substituting the baseline mBERT architecture with XLM-RoBERTa, the study retains the proven ABSA methodology for localized Philippine texts while significantly optimizing the model’s ability to process the complex, informal code-switching of Davao City consumers, as supported by the recent performance benchmarks of Salve and Tubil [45].

2.6 Web Scraping

Web scraping refers to the automated extraction of unstructured data from websites and its conversion into machine-readable formats [49][12]. In sentiment analysis research, it enables large-scale retrieval of user-generated content, such as consumer reviews, that would otherwise be impractical to collect manually [12][16].

Digital data collection methods generally fall into three categories: manual gathering, direct API access, and custom HTML or DOM scraping. API access is preferred where available, as it returns clean, structured data. Mathayomchan and Taecharungroj [34], for instance, drew on Google Maps data to analyze customer experiences across 5,000 businesses, extracting over 935,000 reviews to measure attributes such as food quality, service, atmosphere, and value. However, it has become increasingly difficult to replicate through official channels [34].

Official APIs have grown substantially more restrictive. Chen et al. [12] documented this shift, comparing eight data collection methods and finding that while APIs are straightforward to use, their export limits and associated costs make them impractical for large-scale research. Custom scraping scripts, by contrast, were able to gather sufficient data volume without those constraints [12].

This broader restriction has been described as the "APIcalypse" by Trezza [49]. Examined how platforms banning or severely capping free API access have disrupted large-scale digital research, finding through a survey of digital researchers that API limitations have pushed much of the field toward web scraping as the primary means of maintaining data volume and research viability.

Dongo et al. [16] compared official APIs against custom HTML scraping for real-time social media data collection and found that scraping was considerably faster and more flexible for high-volume retrieval. Crucially, when the scraped text was properly normalized, the two methods produced equivalent analytical results, suggesting that data quality need not be sacrificed for scale [16].

In the context of this study, the standard Google Places API limits extraction to 5 reviews per business, far below the volume needed to fine-tune a deep learning model. Following the approaches documented by Chen et al. [12], Trezza [49], and Dongo et al. [16], this study employs custom HTML scraping to work around that ceiling. The textual noise inherent to scraped data will be addressed through normalization and the preprocessing pipeline described by Maceda et al. [32], ensuring the model receives sufficiently clean input for fine-tuning.

2.7 Synthetic Data Generation

Synthetic text generation creates artificial data using either code-generated data from algorithmic/mathematical models or Large Language Models (LLMs) trained on small seed datasets of real data [53][20]. To deal with the problem of data scarcity, researchers have increasingly looked to LLM driven generation as a workaround. Wang et al. [53] tackled the issue of language models being too dependent on human written instruction data, which is both expensive and limited in quantity and diversity. Their solution was the "Self-Instruct" framework, an iterative setup where a vanilla GPT-3 model was prompted to essentially bootstrap its own dataset by generating new instructions, identifying task types, and creating corresponding input-output instances. After filtering and using over 52,000 purely synthetic instances to fine-tune the model, they saw a 33% absolute performance improvement. That is a pretty significant result, and it effectively matched models trained on large private human datasets [53].

Scaling synthetic generation is one thing, but applying it to structured tasks like ABSA introduces its own set of problems. Monfared et al. [35] found that raw LLM augmentation often struggles to maintain label consistency, meaning the generated sentences frequently fail to preserve the specific aspect level annotations that were actually requested. Their proposed fix was an agentic data augmentation method that uses iterative generation and verification to produce higher quality synthetic training examples. Under a ReAct style architecture, a generator agent first extracted a style policy to produce candidate sentences, while a separate evaluator agent checked that each generated review actually contained the right aspect terms and correct sentiment polarities before it got added to the dataset. Compared to a matched-prompting baseline across three ABSA subtasks, the agentic approach clearly outperformed raw prompting in label preservation [35]. When combined with real data, it also consistently gave stronger performance

gains, particularly for models like T5-Base, which ended up performing at a level more comparable to its more heavily pretrained counterparts [35].

However, efficiency and scalability is considered costly [20]. Relying too heavily on synthetic text introduces what is often called "model collapse," and it is a real concern. Guo et al. [20] looked directly at this risk, asking whether training models on data generated by their predecessors might permanently degrade linguistic richness. They simulated recursive fine-tuning experiments across various natural language generation tasks and applied specialized metrics to track how text structure changed over successive iterations. What they found was a consistent and significant decline in linguistic diversity, specifically in lexical, syntactic, and semantic richness. Models trained excessively on synthetic data eventually converge to uniform behavior and lose the "tails" of the original distribution [20].

Compared to traditional manual annotation, synthetic data generation is the more scalable option for dealing with data scarcity [20] [35]. As Wang et al. [53] showed, synthetic generation can rapidly produce massive datasets from small seed samples. Still, to avoid the linguistic homogenization that Guo et al. [20] warned about, synthetic generation has to be carefully balanced with authentic data rather than used as a straight replacement.

With these findings in mind, this study will use the Self-instruct framework from Wang et al. [53] to augment training datasets, though model collapse as discussed by Guo et al. [20] remains a real boundary that cannot be ignored. Synthetic generation will not replace actual data collection. It will instead serve as a targeted fallback, producing synthetic Bislish and Taglish reviews only for specific aspect-sentiment pairs that stayed underrepresented after manual annotation. Keeping it scoped this way lets the model benefit from more balanced class distributions without losing what makes Davao City consumer reviews distinct in the first place

2.8 Aspect-Based Sentiment Analysis (ABSA)

General sentiment analysis gives a broad picture of consumer satisfaction, but it tends to fall short when reviews are complex, when a single sentence praises one thing and criticizes another in the same breath [13][15][7]. Aspect-Based Sentiment Analysis (ABSA) addresses this by pulling out specific entities or topics and identifying the sentiment attached to each one individually [13]. For a saturated market like the local coffee and matcha industry, where reviews frequently blend Bislish and Taglish to talk about distinct things like taste or service, having a robust ABSA methodology matters if the goal is actually actionable, granular insight.

One of the bigger challenges in applying ABSA to regional service sectors is code-switching, which tends to quietly demolish the accuracy of standard NLP tools [15]. Dionela et al. [15] looked at this problem specifically within Taglish, a low resource, code-switched blend of Tagalog and English that shows up constantly in Filipino consumer reviews. They built a pipeline comparing a traditional rule-based system, a generative LLM, and fine-tuned Gemma-3 models for identifying and extracting both explicit and

implicit aspects. The generative LLM came out ahead across all tasks, which suggests that advanced generative models handle the messy syntactic structures and informal phrasing of Philippine code-switched reviews better than the alternatives [15].

Getting the aspects out is only part of the problem though. Accurately classifying the sentiment and emotional nuance inside code-mixed text is its own challenge [46]. Suciati and Budi [46] tackled this in the context of code-mixed Indonesian reviews, pointing out that standard ternary sentiment, positive, negative, neutral, often lacks the depth needed for nuanced consumer analysis. They compared traditional machine learning algorithms like Decision Trees, Random Forest, and Support Vector Machines against deep learning approaches like GRU and BiLSTM for multi-label classification across restaurant aspects such as food, price, service, and ambiance. The result was that traditional ensemble methods, specifically Random Forest and Extra Trees, actually outperformed the more complex deep learning models on certain tasks and achieved high F1 scores as high as 89.54% for food and 92.65% for service. The takeaway is that for highly localized, code-mixed datasets, model architecture must be matched carefully to the specific aspect being evaluated rather than just defaulting to whatever is most complex [46].

Furthermore, capturing the contextual meaning of localized slang and code-switching also requires more than standard text representation [7]. Amalia and Winarko [7] identified this gap with traditional word embeddings, which struggle to track how the meaning of a word shifts depending on context, particularly in unstructured Indonesian restaurant reviews. Their fix was a hybrid architecture combining a Convolutional Neural Network (CNN) for feature extraction with various contextualized word embeddings, comparing BERT, ELMo, and Word2vec across aspect classification tasks. The ELMo-CNN combination came out on top, which reinforced the idea that pairing deep contextualized embeddings with a solid neural classifier meaningfully improves how well a model reads the nuanced, localized semantics of Southeast Asian consumer reviews [7].

With these findings as a guide, this study will adopt the multi-label aspect framework validated by Suciati and Budi [46] to analyze the kind of unstructured, fragmented feedback that shows up regularly in Davao City café reviews. An XLM-RoBERTa architecture will handle the multi-label classification, isolating and evaluating specific aspect-sentiment pairs. The goal is to make sure that the large volume of complex, code-mixed reviews actually gets converted into structured, usable insights that reflect the operational realities of local coffee and matcha businesses rather than just producing generic sentiment scores.

2.8.1 Aspect Extraction

To identify specific attributes or features from opinionated texts, an Aspect-based sentiment analysis (ABSA) is conducted, where aspect extraction plays a key role. However, unlike traditional sentiment analysis methods, rather than assigning the entire text a single polarity, ABSA identifies a review’s varying aspects along with their respective polarity. Since multiple sentiments can be expressed in a single review, ABSA provides a more detailed understanding of user feedback [36].

Mukherjee and Liu [36] conducted a study and identified the limitations of existing

aspect extraction approaches, particularly those that either extract aspect terms without proper categorization or rely on unsupervised topic modeling, which may produce aspects that are not meaningful or aligned with user needs. Hence, aspect categorization is subjective, as different applications require different categories of the aspects. For that, they introduced a semi-supervised modeling approach for aspect extraction and categorization using user-given seed words. Their models, SAS, and ME-SAS also model sentiment jointly with extraction of aspect terms and grouping them into categories.

In contrast, Poria et al. [42] dealt with the problem of implicit aspects which are neglected by previous methods. They noted that implicit aspects are typically expressed through contextual clues rather than direct mentions, making them more difficult to detect. To this end, they proposed a rule-based approach, common-sense knowledge, and dependency parsing, which extract both explicit and implicit aspects. The implicit aspect clue is determined and mapped to an aspect category with linguistic rules and semantic information. Their research work showed that the detection accuracy of the developed approach is higher than existing techniques.

Suryabrata et al. [47] addressed the challenges of aspect extraction in the case of multilingual and code-mixed scenarios by proposing a transformer-based ABSA framework relying on fine-tuned multilingual models, including mBERT and XLM-RoBERTa. Their major innovation is the sentence-pair classification strategy, which reformulates aspect extraction as a Natural Language Inference (NLI) task. The review sentences were then paired with candidate aspect-sentiment expressions in the framework, enhancing the capture of contextual relations between aspects and sentiments. For evaluation, their samples set of 20,000 aspect-level instances were split into (70%) training, (15%) validation, and (15%) test sets using random sampling without stratification. The sample splits were reused for all models to observe a fair comparison between single-sentence classifiers and their proposed sentence-paired models.

Their results showed that the XLM-RoBERTa achieved an accuracy of 97.20% and an F1-score of 0.9729, while mBERT showed more stability in performance across the validation sets.

Among the reviewed studies, the work of Suryabrata et al. [47] best fits the proposed framework. This, combined with a sentence-pair (NLI) approach, allows for more accurate identification of implicit and explicit aspect-sentiment pairs in a multilingual-code switched environment with a fine-tuned version of mBERT. The approach fits with the study objectives since the information to be analyzed is unstructured, in different languages, and context-specific consumer reviews.

2.8.2 Aspect Taxonomy

Building an aspect taxonomy is one of the more consequential early decisions in any ABSA study because it shapes what the model can and cannot detect downstream [10][9]. Older approaches relied on fixed dictionaries, which tend to miss necessary attributes and break down quickly when the text is informal or syntactically irregular [15]. More recent work has moved toward using Large Language Models to induce taxonomies directly from raw data, without requiring predefined categories or domain-specific seed sets. In a study

In a study, Brady and Islam [10] took this direction when analyzing political advertising content on social media in the 2024 U.S. presidential election. Since the content was highly subjective and shifted rapidly, they built an end-to-end framework that combined unsupervised clustering with prompt-based LLM inference. The model constructed a taxonomy iteratively, without needing seed sets or prior domain knowledge, and the resulting topic labels were more semantically coherent and interpretable than those produced by classical methods [10].

Boughanmi et al. [9] applied a similar approach to food and beverage feedback, working with 20,000 Yelp reviews of Starbucks locations. They tested eight prompt variants against manual human coding and found that the LLM consistently matched human annotators while processing each review. Beyond speed, the extracted attributes showed strong predictive validity for customer ratings, which suggests that generative extraction is not just efficient but substantively accurate [9].

The challenge becomes considerably harder in low-resource and code-switched settings. Dionela et al. (2026) [15] examined this directly using Filipino e-commerce reviews written in Taglish, where standard NLP tools and rigid rule-based systems tend to fail. Their pipeline compared rule-based extraction, a fine-tuned Gemma-3 model, and a generative LLM (Gemini 2.0 Flash). The generative model outperformed the other approaches across all extraction tasks, handling implicit aspects and informal syntax without relying on any predefined lexicon [15].

This study follows the framework used by Dionela et al. [15] for its taxonomy discovery phase. Reviews from Davao City coffee and matcha shops are written heavily in Bislish and Taglish, which means a standard industry taxonomy would likely miss locally relevant concerns like specific ambiance traits or parking availability. Deploying a generative LLM as an unsupervised aspect allows the taxonomy to emerge from the actual language consumers use, rather than from categories decided in advance [15].

2.9 Active Learning Frameworks

Fine-tuning data demands significant amounts of high quality labeled data. When it comes with large volumes of scraped consumer reviews, exhaustive manual annotation quickly becomes a serious bottleneck [37][48][33]. Tharwat and Schenck [48] examined this problem, noting that even with an abundance of free, unlabeled text from digital platforms, the time and cost of manual labeling remain among the biggest obstacles to building supervised learning models. Their survey pointed to Active Learning as a practical workaround. Rather than labeling an entire dataset blindly, AL uses specific query strategies to find the most informative and uncertain data points in the instance space. Focusing human effort on high-value samples achieves strong model generalizability at a fraction of the usual labeling cost [48].

Considering that the feedback is multi-dimensional and complex. Muralidhar and Hulipallal [37] addressed the time and cost constraints associated with using expert human annotators for ABSA on unstructured social media data. They built a Deep Active Learning framework designed to predict aspect-based ratings for smartphones, using un-

certainty sampling to iteratively pick out only the most confusing or informative reviews for human annotation, while a lexicon-based filter removed uninformative neutral statements at the same time. The DAL model hit 92% accuracy, which is a strong result, and it showed that intelligently querying uncertain samples can drastically cut down annotation effort without hurting predictive performance [37].

The same logic extends to fine-tuning modern language models. Margatina [33] explored this in her work on data efficient language models, starting from the observation that not all training examples contribute equally to the learning process even when large amounts are available. By simulating various active learning algorithms, she showed that AL guides training more intelligently by selecting only the most informative unlabeled data points at each iteration. Optimizing data selection this way consistently outperformed random sampling, letting language models reach strong performance with far fewer human labeled instances [33].

These findings directly inform how this study will handle the annotation bottleneck for Davao City consumer reviews. Following the approach demonstrated by Muralidhar and Hulipalled [37], the system will use active learning to selectively surface only the most uncertain Bislish and Taglish samples for manual review. This keeps human effort manageable while ensuring the XLM-RoBERTa model is trained on the most informative data available, which is the central aim of bridging raw web-scraped text and a robust aspect-based classification system [46][15].

2.10 Manual Labeling in Sentiment Analysis

Active learning mechanisms are good at identifying the most uncertain or informative instances in a text corpus, but identifying them is only half the problem [31]. These algorithms cannot actually resolve the ambiguity on their own. What they need, and what every active learning pipeline fundamentally depends on, is a human annotator to provide the definitive ground truth for those flagged edge cases [31]. This is why manual labeling is not just a supplementary step but a necessary one in any sentiment analysis workflow [1][6].

In the Philippine context, where text in localized communications tends to be noisy and heavily context dependent, determining the situational validity of messages requires careful, extensive analysis. Ablazo [1] addressed this by using crowdsourcing through Google Forms, recruiting 250 undergraduate students from Ateneo de Davao University to serve as annotators. The dataset was split into batches, and each instance was independently labeled three times using predefined situational categories. A custom script then checked for annotator agreement, accepting a label only when at least two out of three annotators reached the same conclusion. From a total of 12,054 cleaned instances, the process produced 11,340 entries with a matched majority agreement. Instances where the annotators could not agree were discarded, so only labels that actually had consensus were passed on to the automated classifiers [1].

The question of how to direct that human effort more efficiently is what Li et al. [31] tried to answer. They note that deep learning models need large datasets, but relying

on instance by instance manual labeling makes it hard to balance label quality with cost. Programmatic labeling can scale up, but it produces noisy results. Active learning, on the other hand, struggles with cold start problems when there is not enough labeled data to begin with. To address these limitations together, Li et al. [31] developed the DALL framework, which combines data programming, active learning, and large language models into one integrated labeling system. Rather than annotating random samples, DALL uses active learning strategies like uncertainty sampling, query by committee, and abstain sampling to flag the instances most likely to confuse or conflict with current model predictions. These are then routed to human annotators, who correct the labels and refine the rules [31].

Fine-tuning pretrained language models for Aspect-Based Sentiment Analysis is another difficulty since the annotation needs to be both granular and large in scale. Alturayef et al. [6] point out a specific problem with standard active learning in this context. Algorithms often query instances that are semantically similar to each other, thus, human annotators end up labeling the same kinds of phrasings repeatedly without actually adding new information. To fix this, Alturayef et al. [6] proposed the EASE framework, which builds diversity and data augmentation into the active learning process. Their method calculates contextual similarity scores using Sentence BERT and averages those with model confidence scores, so that the samples flagged for human review are not only uncertain but also meaningfully different from what has already been labeled. After annotation, those edge cases go through Easy Data Augmentation to generate additional training samples synthetically. By filtering out redundant instances before they ever reached human annotators, the framework reduced the total number of manually labeled samples needed by somewhere between 88% and 94% across benchmark datasets [6].

Building on these findings, this study adopts the active learning and data augmentation principles from Alturayef et al. [6] to guide its manual annotation phase. The dataset is made up of highly contextual, code-switched Taglish and Bislish reviews from local coffee and matcha establishments, so exhaustive manual labeling is simply not practical given the scale and variety of the data. By applying the framework’s dual requirement of uncertainty and sample diversity, the pipeline makes sure human annotators are only asked to evaluate complex, non-redundant linguistic edge cases rather than running through similar reviews that do not add anything new [6].

2.10.1 Inter-Rater Reliability

Once the complex, non-redundant edge cases are identified for human review, keeping the resulting annotations consistent becomes its own challenge. Using a majority voting strategy, like the consensus approach Ablazo [1] demonstrated, provides a clear label for subjective and code-switched text, but relying only on simple percentage agreement is often not enough for machine learning applications that need more rigorous validation.

The importance of choosing the right IRR metric for qualitative annotations is a topic the literature has addressed fairly directly. El Dehaibi and MacDonald [17] looked at IRR measures for qualitative annotations in machine learning datasets and found that running an IRR check before training provides a necessary internal validity step. They

explain that Cohen’s Kappa corrects for chance agreement between exactly two raters on categorical items, but once a study involves three or more raters, Fleiss’ Kappa becomes the appropriate metric.

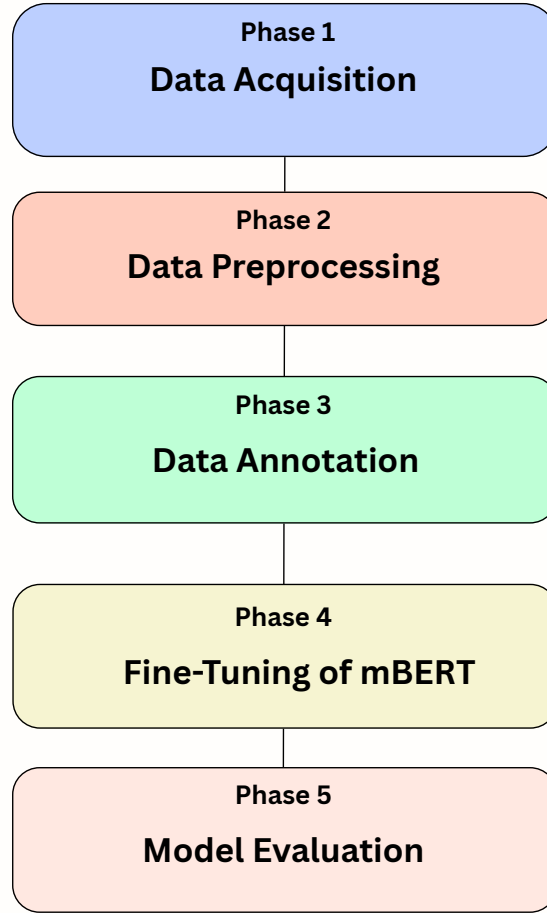
Chamid et al. [11] show what Cohen’s Kappa looks like applied to a more demanding annotation scenario. Their study evaluated labeling consistency for ABSA using 4,307 Indonesian marketplace reviews, where two expert linguists independently labeled aspect detection, sentiment classification, and aspect class. The resulting Kappa values were 0.909 for aspect detection, 0.893 for sentiment classification, and 0.971 for aspect class, all falling in the almost perfect agreement range. These scores confirmed that the annotated dataset was reliable enough to actually train accurate classifiers on.

When annotation design scales up to three or more raters, Fleiss’ Kappa is discussed. Krušić [29] illustrates this in building a sentiment annotated corpus of historical texts for NLP, where three semi-expert annotators categorized sentences into positive, negative, neutral, and mixed sentiment. Fleiss’ Kappa across the three raters came out to 0.405, which falls in the fair to moderate range. A majority vote requiring at least two out of three annotators to agree was used to finalize labels, and the overall majority vote agreement reached 92.5%. These results were considered sufficient to establish the annotations as a reliable enough corpus for fine-tuning machine learning models.

2.11 Theoretical Frameworks

2.11.1 Multilingual Transformers

This method of Multilingual Transformers is grounded in the methodology of Maceda et al. [32]. Their study highlights the need for multilingual models to process code-mixed data, demonstrating that traditional approaches are insufficient for analyzing text that blends languages such as English and Filipino due to inconsistencies in grammar, vocabulary, structure, and context. By implementing a fine-tuned multilingual Bidirectional Encoder Representations from Transformers (mBERT), they were able to capture contextual relationships across multiple languages. The sequential process for this framework is depicted in Figure 1. This approach establishes that multilingual transformers are well-suited for handling linguistically diverse datasets, supporting their application in studies involving mixed-language inputs. Similarly, Patil et al. [40] emphasized that transformer-based approaches consistently outperform traditional feature-based models when handling noisy text. Therefore, multilingual transformers are considered an optimal modeling framework for analyzing unstructured, code-mixed consumer reviews where linguistic variability is prevalent.

Figure 1: *Multilingual Transformers*

Phase 1: Data Acquisition The framework begins by collecting unstructured, code-mixed textual data from multiple social media platforms, such as Facebook, Twitter, and YouTube. Scraper libraries were utilized to systematically retrieve relevant posts, tweets, and comments using predefined keywords and sources [32].

Phase 2: Data Preprocessing The noisy and inconsistent nature of the code-mixed dataset necessitated preprocessing to clean and standardize the data. Duplicate entries were removed using URL-based identification, and irrelevant content was filtered manually. Minimal preprocessing techniques were applied, including the correction of informal spelling, normalization of special characters, removal of unnecessary whitespace and punctuation, and conversion of emojis into their textual representations. This approach preserves contextual meaning while improving data consistency [32].

Phase 3: Data Annotation From the preprocessed dataset, data points were manually labeled and validated by experts. Each instance was classified into three sentiment

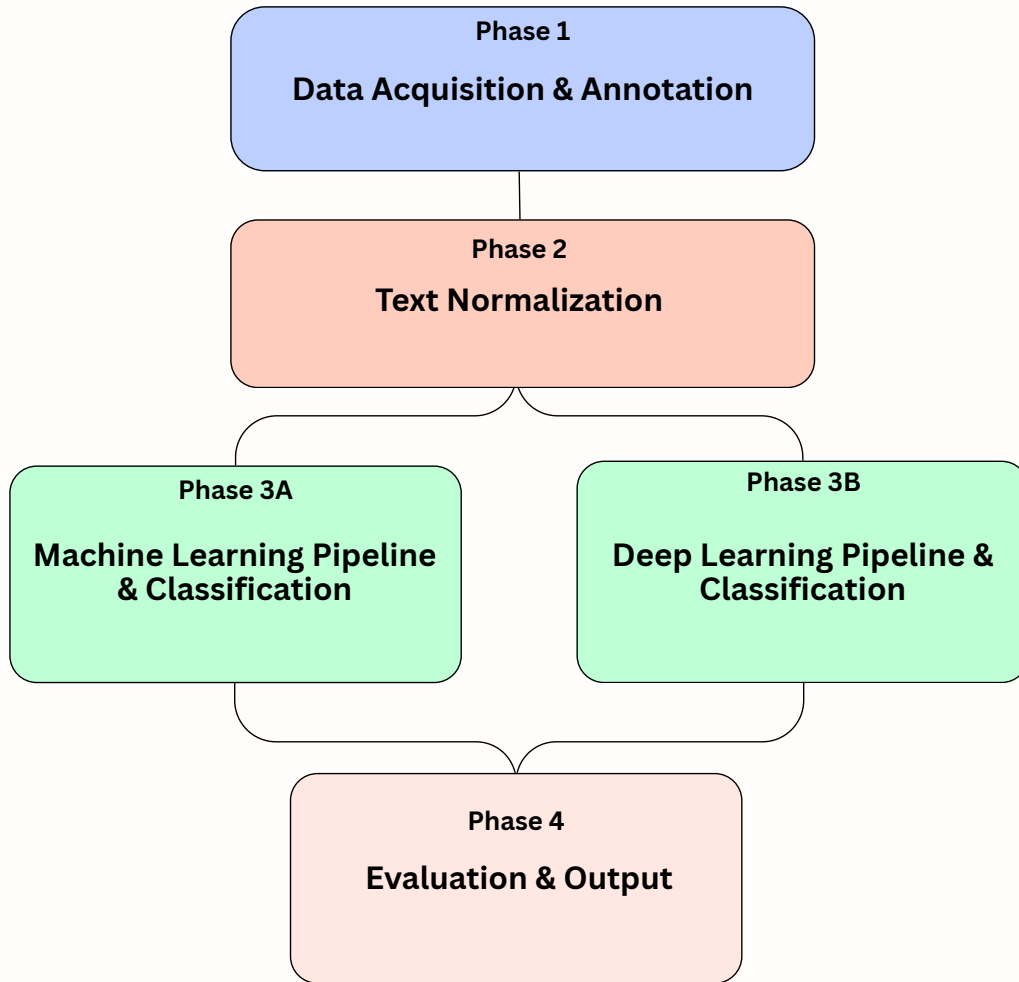
categories: neutral, negative, and positive, encoded as 0, 1, and 2, respectively. The annotated dataset serves as the ground truth for training and evaluating the sentiment classification model, following established annotation practices [32].

Phase 4: Fine-Tuning of mBERT The framework utilizes a fine-tuned Multilingual Bidirectional Encoder Representations from Transformers (mBERT) model for sentiment classification. Using transfer learning, the model will leverage pre-trained global linguistic knowledge and adapt it to the specific variations in the mode-mixed dataset. Built on the Transformer architecture, mBERT captures contextual relationships across multiple languages, making it suitable for code-mixed data. The model processes input sequences using special tokens such as [CLS] for classification and [SEP] for segmentation, where the final hidden state of the [CLS] token is used for prediction. Fine-tuning was performed using task-specific parameters, including learning rates ($2e - 5$, $3e - 5$, $5e - 5$), batch sizes (16 and 32), and four training epochs. A maximum sequence length of 160 tokens was applied, and class weights were used to address class imbalance [32].

Phase 5: Model Evaluation The model’s performance was evaluated using accuracy and F1-score metrics to assess its effectiveness in classifying sentiment in code-mixed text. The best-performing model configuration was then used to predict sentiment labels for the remaining dataset, enabling the analysis of overall sentiment distribution. This evaluation ensures the reliability and robustness of the framework in handling multilingual and unstructured data [32].

2.11.2 Aspect-based sentiment analysis

The method of Aspect-based sentiment analysis is grounded by the methodology by Suciati and Budi [46]. Their research addresses computational complexities of processing code-mixed (Indonesian-English) restaurant reviews by proposing a dual-pipeline classification architecture, as illustrated in Figure 2. Since standard single-language models fail to capture the nuances of bilingual reviews [22], their framework compares the potency of traditional Machine Learning (ML) models against advanced Deep Learning (DL) networks in performing multi-label classification for both sentiment and emotion.

Figure 2: *Aspect-based sentiment analysis*

Phase 1: Data Acquisition and Annotation The framework is based on collecting unstructured review data that contains naturally mixed formal language and informal code-switching. The data then goes through the annotation processes using crowdsourcing. Multiple annotators label the text based on predefined guidelines, and when annotators reach a disagreement on how something should be labeled, a majority vote settles the call [46].

Phase 2: Text Normalization (Pre-processing) As code-mixed social media text is highly noisy and inconsistent, the framework implements an extensive eight-step preprocessing layer. Abbreviations are corrected using self-developed bilingual dictionaries, while word stemming is handled by two dedicated tools: the Snowball Stemmer for English and the Sastrawi Stemmer for Indonesian. Emoticons are also converted into their text equivalents to ensure that meaningful emotional cues are preserved throughout the cleaning process [46].

Phase 3: Dual Classification Pipelines

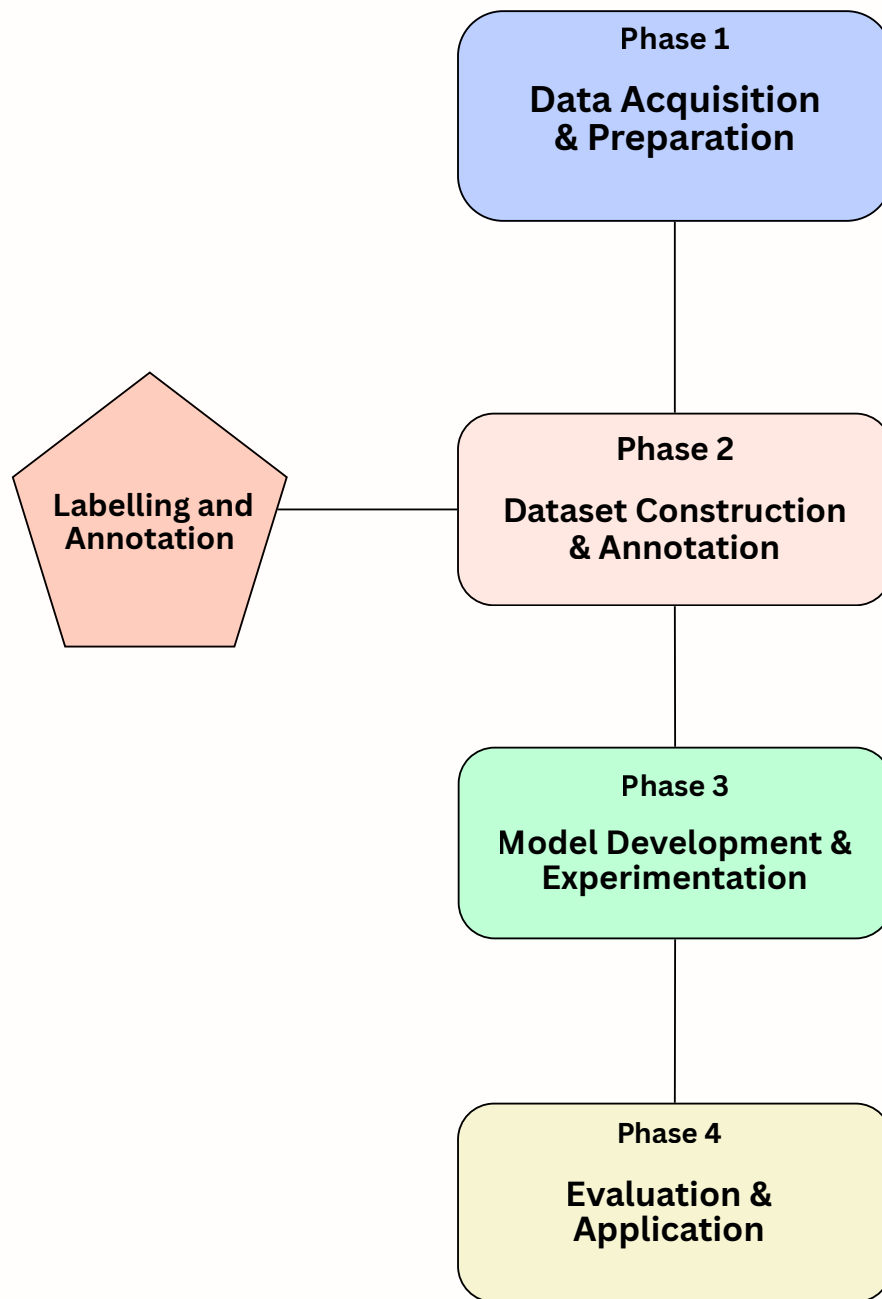
- *Machine Learning Pipeline:* The Machine Learning pipeline relies on traditional feature extraction methods through N-gram language models, such as unigrams, bigrams, and a combination of both, paired with Chi-Square for feature selection. Since aspect-based classification often involves reviews that address multiple targets at the same time, the framework applies problem transformation methods such as Binary Relevance, Classifier Chain, and Label Powerset to adapt standard algorithms including SVM, Random Forest, Decision Tree, and Extra Tree into multi-label classifiers [46].
- *Deep Learning Pipeline:* The deep learning pipeline takes a different approach by eliminating the need for manual feature engineering. Instead, it constructs custom word embeddings using the Skip-gram method to capture the semantic relationships present in code-mixed text. Deep learning architectures, specifically BiLSTM and GRU, are then deployed using a sigmoid activation function and binary cross-entropy loss to directly predict multiple aspect-sentiment pairs at the same time [46].

Phase 4: Evaluation and Output The performance of both pipelines is thoroughly tested using a 10-fold cross-validation technique. The framework evaluates each model based on its F1-Score, measuring how accurately it was able to isolate and classify sentiment and emotion across four distinct business dimensions: food, price, service, and ambience [46].

2.11.3 Aspect Extraction

This framework draws from Natural Language Processing, transfer learning, and Aspect-Based Sentiment Analysis. The core problem with traditional sentiment analysis is that it reduces an entire review to a single polarity label, and that is simply too blunt for user-generated content [47]. In the study, hotel guests routinely praise one aspect of their stay while criticizing another in the same sentence. Recovering that granularity requires Aspect Extraction. It is the process of identifying which specific service attributes a reviewer is actually referring to [47].

The study uses mBERT and XLM-RoBERTa, two multilingual transformer architectures, to handle the code-mixed language that commonly appears in reviews from multilingual settings. Since no locally annotated dataset exists for this domain, zero-shot learning was adopted. The ABSA task is reformulated as a Natural Language Inference problem, which lets the models assess sentiment without needing labeled training examples from the target domain [47]. Figure 3 outlines the overall workflow and the structural phases of the study.

Figure 3: *Aspect Extraction*

Phase 1: Data Acquisition and Preparation Any NLP pipeline depends on the quality and representativeness of its corpus. For this study, over 100,000 reviews were collected from TripAdvisor and Google Reviews across 43 five-star hotels in Jakarta. The challenge at this stage is that user-generated content is unstructured, inconsistent, and often multilingual, with some reviewers switching between languages mid-text [47].

- *Collection and Preprocessing* Covers data collection and preprocessing. Duplicate entries and empty strings are removed to reduce model bias and prevent overfitting. A language filter then isolates reviews in Indonesian, English, and code-mixed text. This filtering decision was deliberate: mBERT and XLM-RoBERTa were pre-trained on multilingual corpora, and keeping the input data within those language boundaries ensures the models are operating within their intended distribution [47].

Phase 2: Dataset Construction and Annotation This is where Zero-Shot Learning and Aspect Extraction are present. A major hurdle in ABSA is the "cold-start" problem: the lack of massive, human-annotated datasets for specific domains (like Indonesian hotel reviews). This phase leverages the theory of Zero-Shot Transfer Learning, in which a model pre-trained on general tasks is used to classify unseen data without requiring domain-specific fine-tuning. Crucially, it relies on aspect extraction as the prerequisite mechanism to break down monolithic blocks of review text into granular, analyzable targets before any sentiment can be measured [47].

- *Zero-Shot Labeling and Aspect Extraction:* multilingual zero-shot classifier acts as a proxy annotator. The indispensable first step is aspect extraction, where the model automatically scans the raw text to identify what the user is talking about, categorizing the text into seven predefined aspects (Cleanliness, Facilities, Food, Location, Price, Room, and Service). Because a single review might discuss multiple aspects simultaneously. In the dataset, sentiments were isolates each extracted review-aspect pair into its own unique instance (generating 352,720 records). Only after these target aspects are successfully extracted and separated can a subsequent zero-shot pass accurately determine the specific sentiment (Positive, Negative, Neutral) attached to them [47].
- *Sampling and Splitting:* To optimize computational efficiency while preserving statistical significance, a representative sample of 20,000 aspect-level instances is extracted and rigorously partitioned (70% Train, 15% Validation, 15% Test) to ensure the model can be evaluated on strictly unseen data [47].

Phase 3: Model Development and Experimentation This phase is where the core computational work happens. Rather than treating ABSA as a standard classification task, the study reformulates it as a Natural Language Inference problem [47].

- *Sentence-Pair Construction and Tokenization:* Each review is restructured into a sentence pair. The review text serves as the premise, and a constructed auxiliary sentence. The model then predicts whether the premise entails the hypothesis. Framing the task this way gives the transformer explicit structural context about the aspect and sentiment orientation being evaluated. Tokenization is handled by subword algorithms appropriate to each model. WordPiece for mBERT and SentencePiece for XLM-RoBERTa, with the other being better equipped for the morphological complexity and code-switching present in the corpus, morphological variation and code-switching [47].
- *Training and Fine-Tuning:* One persistent problem with review datasets is se-

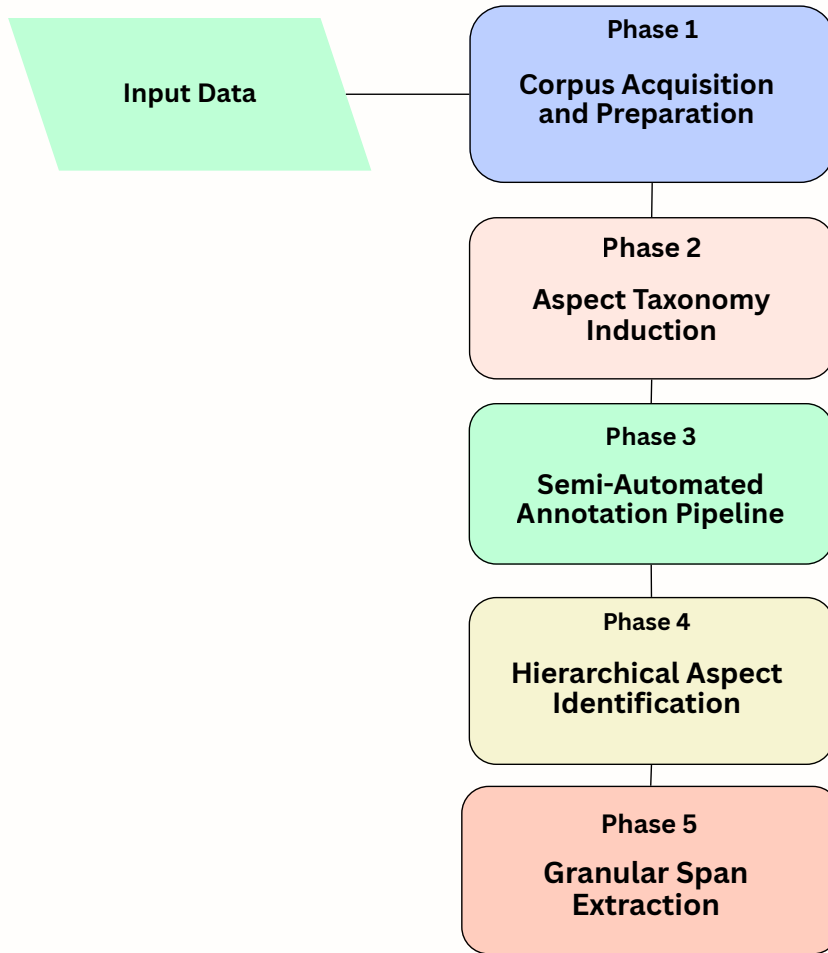
vere class imbalance, as positive reviews typically account for more than 90% of the data. A model trained on distribution can achieve deceptively high accuracy by defaulting to positive predictions while failing almost entirely on negative and neutral cases. Rather than correcting this through oversampling, which carries a real risk of overfitting, the framework applies Weighted Loss Functions under a cost-sensitive learning approach. Misclassifications of minority class instances are penalized more heavily during training, pushing the model to learn the distinguishing patterns of negative and neutral reviews rather than treating them as statistical noise [47].

Phase 4: Evaluation and application The final phase connects model outputs to practical use within a hotel management context.

- *Evaluation:* On imbalanced datasets, standard accuracy is an unreliable measure of model performance. A classifier that predicts positive sentiment for every review would exceed 90% accuracy on this corpus while failing entirely to detect negative or neutral cases. For this reason, the framework uses the F1-score as its primary evaluation metric, which computes the harmonic mean of Precision and Recall. Precision measures how many of the model’s predicted positives are correct, while Recall measures how many of the model’s predictions are correct. Confusion matrices and error analyses supplement these scores by identifying which linguistic structures consistently cause misclassification, providing diagnostic depth beyond what aggregate metrics alone can offer [47].
- *Application and Visualization:* Deploys the better performing architecture, XLM-RoBERTa, across the approximately 76,000 reviews that were not used in training or evaluation. The resulting sentiment predictions are mapped into visualizations including heatmaps and correlation charts. This translation step is what makes the model outputs usable in practice so that hotel management can identify which service aspects are consistently underperforming across reviews without having to interpret raw model inferences directly [47].

2.11.4 Aspect Taxonomy

This research builds on two related ideas: the Principle of Cross Lingual Semantic Resilience and the Theory of Generative Attribute Induction. Dionela et al. [15] incorporated both into their Aspect-Based Sentiment Analysis framework [15]. The claim is that generative language models can process messy, code-switched consumer feedback without relying on predefined word lists. Dionela et al. [15] points out that traditional Natural Language Processing fails in low-resource, mixed-language environments [15]. Their study of Philippine e-commerce reviews shows the problem clearly. Rule based models and small fine tuned models struggle with code-switched Taglish since they depend on keyword matching [15]. When consumers use informal language or imply an aspect without stating it directly, the system misses it. For instance, "It took a week to arrive" signals a delivery problem, but the keywords might not be obvious [15]. Refer to Figure 4.

Figure 4: *Aspect Taxonomy*

Phase 1: Corpus Acquisition and Preparation. Raw, unstructured code-switched consumer reviews are processed to create a baseline corpus [15]. The dataset contains 10,510 Taglish reviews from Google Maps and Shopee Philippines. Human annotators labeled both general and specific aspect categories [15].

Phase 2: Aspect Taxonomy Induction Rather than starting with predefined categories, this phase builds a Hierarchical Aspect Framework by combining four approaches: Open Coding, Latent Dirichlet Allocation for topic modeling, BERTopic for multilingual embeddings, and Generative LLMs to integrate the results [15]. The outcome was a taxonomy with 4 General Categories (Product, Delivery, Price, Service) and 21 Specific Aspects derived from the actual corpus [15].

Phase 3: Semi-Automated Annotation Pipeline To annotate implicit aspects at scale, human researchers first labeled 67 reviews as a reference set, achieving a Fleiss Kappa of 0.691. This reference set was then used as few-shot examples for Gemini 2.0 Flash, which automatically annotated the remaining reviews to create a

training dataset [15].

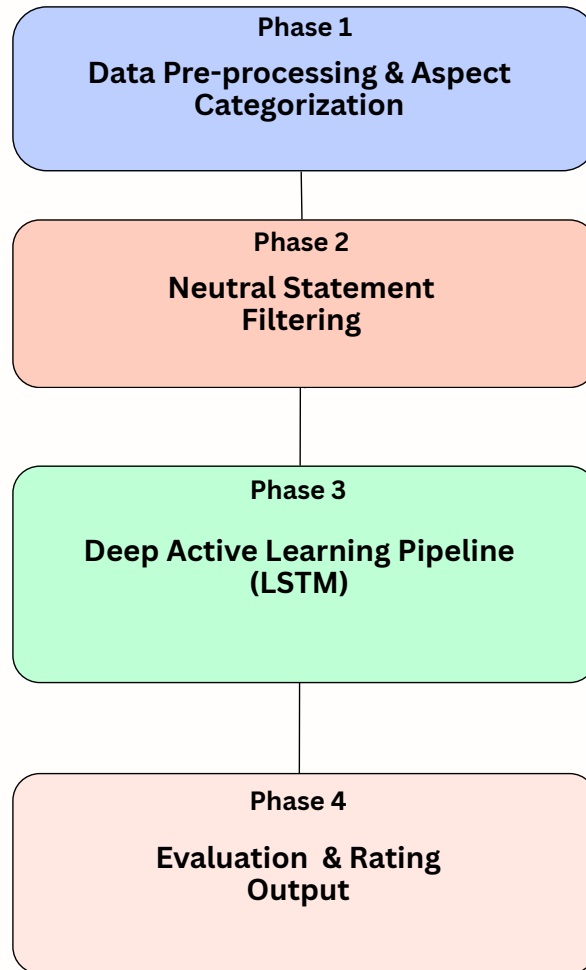
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Phase 4: Hierarchical Aspect Identification The framework uses conditional classification: broad domains like Product or Service are identified first, and these results determine which specific aspect classifiers run next. It was then compared to four approaches. A Rule Based system, Gemini 2.0 Flash, and two versions of Fine Tuned Gemma 3 (1B) [15]. The Generative LLM achieved a Macro F1 of 0.91, outperforming the others. The smaller fine tuned models had difficulty with noisy syntax and imbalanced classes in code-switched text [15].

Phase 5: Granular Span Extraction The system extracts exact text spans for the identified aspects, mapping code-switched vocabulary to the taxonomy. LLMs extracted numerical concepts like price accurately, but struggled with Over Extraction and Partial Matches when faced with variations in Taglish Product and Service terminology [15].

2.11.5 Active Learning

This method of Active Learning is grounded in the methodology established by Muralidhar and Hulipalled [37]. Their research addresses the high cost and time needed to manually label large social media datasets by proposing a Deep Active Learning pipeline as depicted in Figure 5 To reduce misclassification of objective and neutral statements, their framework combines lexicon-based techniques with human judgment to distinguish positive from negative sentiments. Text normalization, aspect categorization, and a Long Short-Term Memory (LSTM) network form the core of this pipeline. The model queries human experts only when classification confidence falls below a set threshold, resulting in efficient, aspect-specific sentiment predictions for mobile application reviews [37].

Figure 5: *Active Learning*

Phase 1: Data Pre-processing and Aspect Categorization Raw social media text presents several preprocessing challenges, such as misspellings, informal jargon, and hashtags that merge multiple words together. The framework addresses these using Ekphrasis, a tool that segments compound tokens, corrects spelling errors, and applies lemmatization to reduce words to their root forms without losing contextual meaning. Emojis and emoticons are converted to their text equivalents before the cleaned text is split into smaller phrases along punctuation and conjunction boundaries. Each phrase is then assigned to a product aspect such as "battery" or "camera" using predefined keyword dictionaries [37].

Phase 2: Neutral Statement Filtering Neutral, non-opinionated statements may distort classifier training if left in the dataset. To filter these out, the framework scores each text entry using a composite lexicon built from VADER, AFINN, and SentiWordNet. Since domain-specific language can complicate automated scoring "heavy" carries negative sentiment in a phone review but registers as neutral in a general lexicon. Then, a human annotator reviews the 100 most frequent words

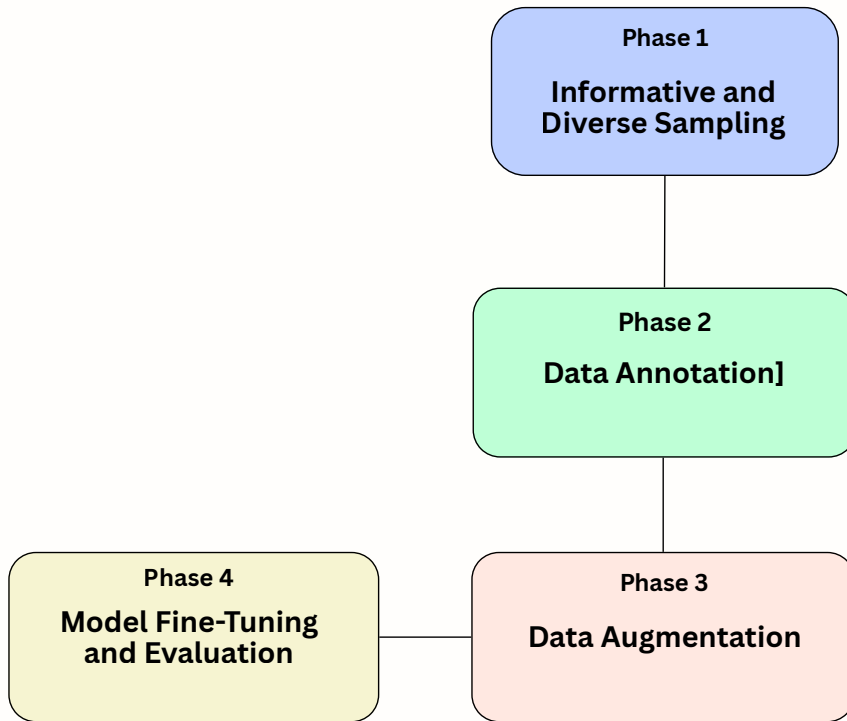
flagged as neutral. Words identified as false neutrals are reassigned sentiment polarities and reintegrated into the lexicon. The dataset is then recorded, and remaining neutral entries are discarded, leaving a training set composed entirely of opinionated samples [37]. While Muralidhar and Hulipalled [37] utilized standard lexicons for this filtering phase, this study’s conceptual framework adapts this approach to accommodate localized code-switching.

Phase 3: Deep Active Learning Pipeline The framework of the specified study uses an LSTM-based Active Learning strategy that selectively involves human input. A human annotator manually labels an initial batch of 5,000 reviews to train the model. The LSTM then processes the next 5,000 reviews: predictions made with at least 75% confidence are accepted automatically, while those falling below this threshold are routed back to the annotator for manual labeling. Each newly labeled sample is used to retrain the model, progressively improving its classification boundaries while keeping total annotation effort low [37].

Phase 4: Evaluation and Rating Output Model performance is evaluated against standard baselines, including SVM, using precision, recall, and F-measure. Beyond classification accuracy, the framework converts binary sentiment predictions into a 1-to-10 aspect-level rating scale, which is calculated by dividing the count of positive reviews by the total number of reviews for each aspect, thus producing a metric directly usable in commercial product evaluation [37].

2.11.6 Manual Labeling in Sentiment Analysis

This method of optimizing data selection for Aspect Based Sentiment Analysis (ABSA) draws from the methodology of Alturayef et al. [6]. Their study identifies manual annotation as a serious bottleneck when fine tuning pretrained language models for granular sentiment tasks. To address this, they proposed the Enhanced Active learning framework for Aspect-based Sentiment Analysis (EASE), demonstrating that filtering redundant data before it reaches human annotators can significantly reduce labeling costs without harming model accuracy [6]. The framework works by combining uncertainty sampling, sample diversity, and targeted data augmentation so that every manually labeled instance actually contributes something useful. The full sequential process is illustrated in Figure 6.

Figure 6: *Manual Labeling in Sentiment Analysis*

Phase 1: Informative and Diverse Sampling The framework starts by evaluating a large pool of unlabeled text. Rather than picking instances at random for human review, the algorithm computes two specific metrics to find the most valuable data points. The first is model uncertainty, which surfaces sentences the model finds ambiguous. The second is a contextual similarity score computed using Sentence-BERT, which checks whether a candidate sample is meaningfully different from the selected sample. Averaging these two scores together, the framework flags only the instances that are both uncertain and linguistically distinct from one another [6].

Phase 2: Data Annotation The informative and diverse instances identified in Phase 1 are then sent to human annotators for manual labeling. Since the algorithm removed semantically similar reviews, annotators spend their time on genuinely unique and complex patterns rather than variations of the same thing. This targeted approach reduces the total number of samples that need manual labeling by up to 94% compared to random sampling [6].

Phase 3: Data Augmentation To get more out of the newly annotated data and to handle potential class imbalances, the labeled instances go through Easy Data Augmentation (EDA). This step generates additional training samples from the seed data, which expands the overall training corpus while preserving the contextual integrity of the edge case reviews that were originally selected [6].

Phase 4: Model Fine-Tuning and Evaluation The final dataset which consists of a diverse manually annotated core supplemented by the augmented synthetic samples, is used to fine-tune the pretrained language model for ABSA. The model is then tested on benchmark datasets to confirm that classification accuracy holds up even with a drastically reduced annotation effort [6].

2.11.7 Synthetic Data Generation

To address data scarcity and class imbalance, the introduced study adopts the Self-Instruct framework developed by Wang et al [53]. The procedural steps of which are shown in Figure 7. The approach uses a pre-trained Large Language Model to generate training data from a small set of human-authored seed examples, expanding the dataset substantially without additional manual annotation.

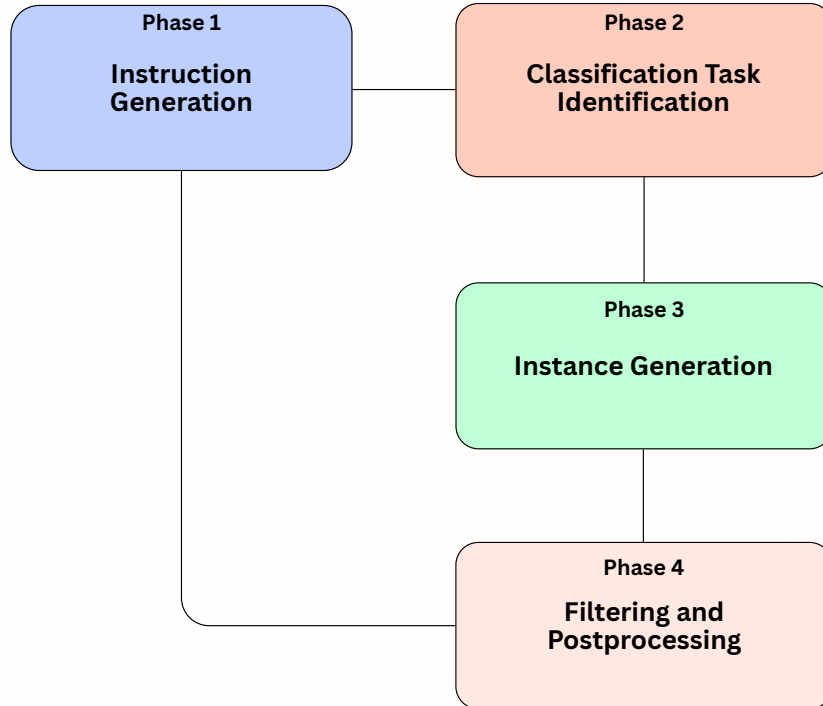


Figure 7: *Synthetic Data Generation*

Phase 1: Instruction Generation The process starts with a Task Pool containing a

small set of human-written seed examples. The LLM draws from these to generate new task instructions recursively, using existing examples as context for each subsequent generation [53].

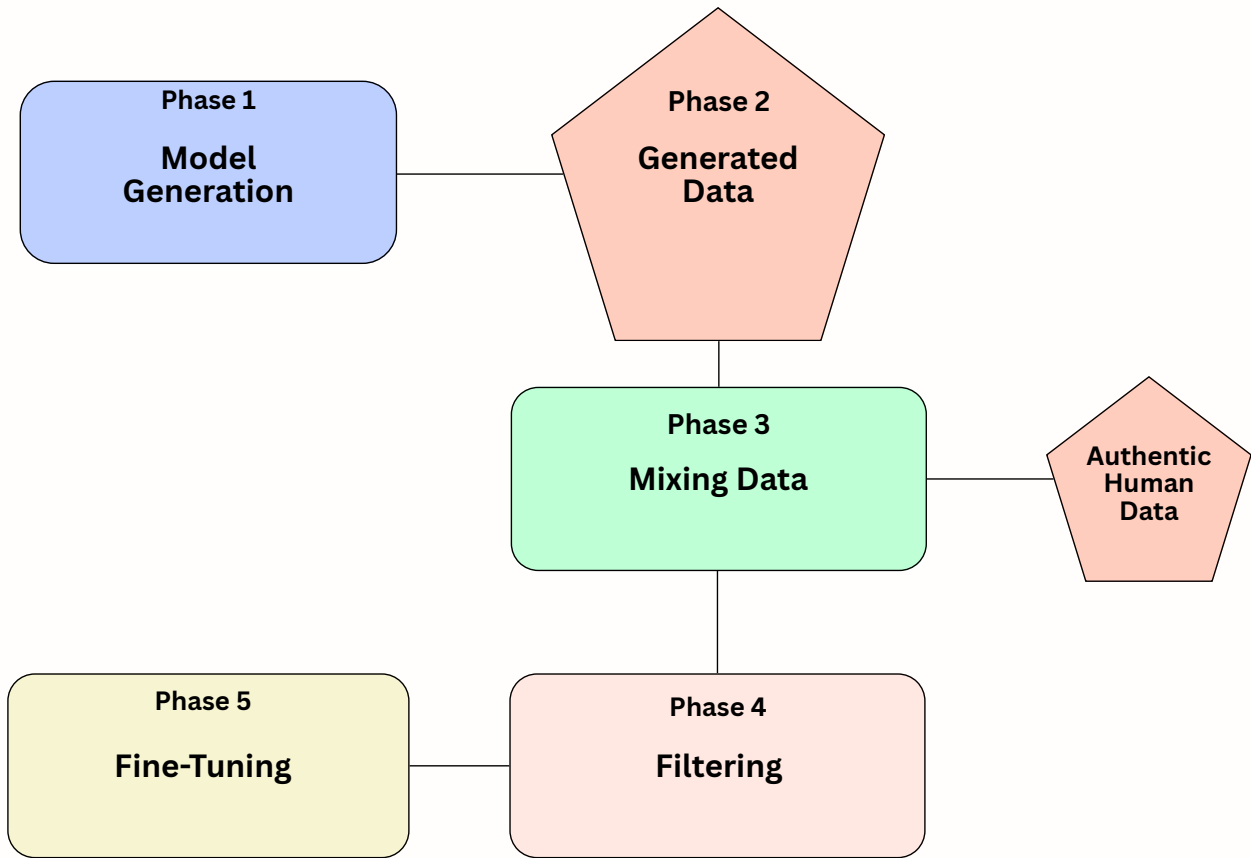
Phase 2: Classification Task Identification The model determines whether it is a classification task, such as assigning sentiment polarity to a review, or a non-classification task. This distinction controls how the input and output are structured in the next phase [53].

Phase 3: Instance Generation Using the instruction and its identified task type, the model generates a synthetic input-output pair. For classification tasks, the input is a synthetic text review, and the output is a set of aspect and sentiment labels. The structure differs for non-classification tasks based on what the instruction specifies [53].

Phase 4: Filtering and Postprocessing Before entering the Task Pool, generated instructions are checked for redundancy and validity. Instructions with a ROUGE-L overlap of 0.7 or above against any existing pool entry are discarded, including entries the system cannot parse. The remaining instances are added to the pool and become available as in-context examples for future rounds, growing the dataset over successive iterations [53].

2.11.8 Constraint of Synthetic Data: Acknowledging Model Collapse

While the study Self-Instruct framework by Wang et al. [53] establishes the methodology for generating synthetic data. Scaling the dataset effectively requires managing the inherent properties of AI-generated text [20]. Preserving the code-switched syntax of localized dialect reviews shapes how the augmentation strategy must be designed. Figure 8 visualizes the framework for mixing synthetic and authentic data to mitigate these constraints. Thus, the theoretical framework of Guo et al. [20] demonstrates that training models on synthetic outputs causes a persistent decrease in linguistic diversity, a phenomenon that cannot be fully cured even when mixed with authentic human data.

Figure 8: *Synthetic Data Causing Model Collapse*

Phases 1 and 2: Generation and Diversity Loss The cycle begins with a language model generating a batch of synthetic text. Guo et al. [20] find that synthetic data consistently underrepresents the lexical, syntactic, and semantic range of human-authored text. This deficit compounds across generation rounds, producing diversity loss even in tightly constrained tasks.

Phase 3: Data Mixing To reflect a realistic training scenario, the framework introduces a mixing phase in which synthetic outputs are combined with a fixed proportion of human-authored data [20].

Phases 4 and 5: Persistent Degradation After filtering and fine-tuning the subsequent model, Guo et al. [20] show that the added human data produces only marginal gains in diversity. The core finding is that model collapse is not a problem mixing alone can solve. Diversity loss accumulates across iterations. Hence, reliance on synthetic data still causes a persistent, cumulative decline in linguistic diversity, eventually leading to “*model collapse*” [20].

The findings of Guo et al. [20] establish that mixing human and synthetic data reduces diversity loss but does not eliminate it. This constraint directly shapes the augmentation strategy used. Davao City Coffee and Matcha shop reviews are characterized by code-switched Bislish and Taglish phrasing that is locally specific and difficult to replicate synthetically. Generating large volumes of synthetic data for this corpus would degrade the lexical and syntactic patterns the model is trained to classify. Therefore, synthetic data generation implemented through Wang et al.’s [53] Self Instruct Framework will serve as a fallback safety net rather than a primary data source. It is applied only when a specific aspect-sentiment class has too few real samples to be usable, such as when a given aspect yields no negative reviews. Outside these cases, training data comes exclusively from human-annotated reviews. Consequently, preserving the linguistic character of the corpus and maintaining class distribution for model training.

To effectively analyze code-switched consumer reviews, this research integrates the Multilingual Transformers framework with the selective querying principles of Deep Active Learning, the unsupervised discovery of an Aspect Taxonomy, and the structural granularity of Aspect Extraction. Traditional feature-based pipelines relying on N-grams and algorithms like SVM will be excluded, as the literature consistently demonstrates that transformer-based architectures outperform them when processing noisy, unstructured text [40] [32] [54]. While the foundational Aspect-Based Sentiment Analysis (ABSA) framework established by Maceda et al. [32] utilized mBERT, this study implements XLM-RoBERTa as the primary classification engine, given its superior capacity to capture complex contextual relationships and resolve ambiguities in highly code-switched Southeast Asian languages [45].

Crucially, since traditional sentiment analysis reduces an entire review to a single polarity label, it is often too blunt for complex user-generated content where consumers frequently praise one attribute while criticizing another in the same sentence [47]. To address this, the framework first incorporates generative Aspect Taxonomy Induction [15] to discover localized operational dimensions from the raw text. This serves as a prerequisite phase for Aspect Extraction. Thus, ensuring the review text is broken down into granular, analyzable targets governed by a data-driven master dictionary rather than assumed categories. By identifying these specific service attributes and reformulating the classification task as a Natural Language Inference problem [47], the model can isolate and evaluate individual review-aspect pairs efficiently.

However, fine-tuning the classification engine requires a large volume of ground truth labels, and getting those labels manually is where the process tends to slow down considerably. To work around this, the methodology draws on the principles of the Enhanced Active learning framework for Aspect based Sentiment Analysis [6]. Instead of labeling everything or pulling samples at random, the pipeline uses uncertainty and sample diversity metrics to identify only the reviews that are genuinely ambiguous and linguistically distinct from one another. By narrowing the annotation queue down to these complex, non redundant cases, human effort gets directed toward the instances that actually matter rather than being spread thin across data that would not meaningfully improve the model [6].

To address data scarcity within this pipeline while keeping model collapse in check, the framework also adopts the Self-Instruct methodology for synthetic data generation [53]. That said, unconstrained reliance on AI-generated text causes a persistent, cumulative decline in linguistic diversity that mixing with authentic data does not fully fix [20], so bulk synthetic generation is deliberately avoided. Synthetic data is positioned strictly as a targeted fallback for underrepresented aspect-sentiment classes, not a primary source.

Taken together, these approaches form a comprehensive framework that draws on XLM-RoBERTa’s robust classification capabilities, generative Aspect Taxonomy discovery, validated manual labeling driven by Fleiss’ Kappa, and constrained synthetic augmentation. The literature supports this combination as a highly effective way to balance class distribution needs and minimize human annotation costs, all while preserving the complex, code-switched syntax that makes localized Bislish and Taglish reviews worth studying.

Chapter 3

Conceptual Framework

The study’s conceptual framework will be described using the Input-Process-Output (IPO) model, as illustrated in Figure 9, which maps the transformation of unstructured, multilingual consumer reviews into actionable market intelligence. The IPO model illustrated the flow of data from raw collection to final classification. Emphasizing the combined application of web scraping, active learning, synthetic data augmentation, and transformer-based sentiment analysis, thus, effectively addressing the limitations of traditional review evaluation in localized, code-switched domains.



Figure 9: *IPO Chart (Input-Process-Output)*

3.1 Input Phase

This defines the foundation for the data requirement. The primary data source consists of unstructured consumer reviews of local coffee and matcha businesses in Davao City. These reviews are gathered through scraping from Google Maps reviews. Its linguistic complexity is one of its unique characteristics. It is multilingual and features heavy code-switching among English, Tagalog (Taglish), and Bisaya (Bislish) [38].

3.2 Process Phase

This outlines the machine learning and data engineering pipeline for handling the input data.

1. **Custom HTML Scraping:** A sufficient volume of reviews is needed, and this necessarily requires bypassing the strict extraction limit of only 5 instances imposed by the standard Google Places API [19]. Data is gathered through custom web scraping scripts [51].
2. **Normalization and Cleaning:** Noisy, unstructured text undergoes preprocessing to correct informal spelling, normalize special characters, and convert emojis to textual representations. Thus, ensuring clean and consistent data for model ingestion [46].
3. **Generative Aspect Taxonomy Induction:** Rather than relying on assumed, static industry categories, generative Large Language Models (LLMs) are used to discover recurring themes and implicit operational dimensions specific to the regional Davao City consumer context (e.g., [Wi-Fi Reliability], [Parking Availability]). This establishes a data-driven master dictionary [15].
4. **Aspect Extraction:** Because reviews frequently contain mixed sentiments directed at different attributes, traditional document-level classification is insufficient. In this stage, the model isolates specific review-aspect pairs (e.g., Taste, Price, Service, Ambiance) from the unstructured text. This serves as a prerequisite mechanism to break down complex sentences into granular, analyzable targets before any sentiment is evaluated [47].
5. **LLM Synthetic Augmentation:** To catch class imbalances early in the pipeline, a data sufficiency threshold checks the class distribution across the isolated aspect dimensions right after extraction. Rather than relying on a fixed, arbitrary target, which often lacks empirical grounding in specific ABSA contexts, the threshold is determined empirically instead. Using a progressive sampling approach [44], a small subset of authentic data is trained incrementally. The model is first fine tuned on an initial batch of 50 instances, then expanded to 100, and then to 150. After each batch, the F1-score is evaluated and plotted to build a learning curve. This process reveals how much data the model actually needs to learn the patterns of a specific category. The point where adding more data stops producing meaningful gains, where the curve levels off, becomes the minimum viable threshold for that class [23][41].

To prevent model collapse [20], this threshold works as a dynamic ceiling rather than a fixed cutoff. If an underrepresented category falls below its empirically established baseline, a generative LLM (Gemini 3) is triggered to synthesize additional samples. Generation continues only until the class either reaches the empirical threshold or matches the median frequency of the naturally occurring classes in the corpus [37], whichever comes first. This keeps augmentation from running unchecked and limits it to a regulated fallback role. If all classes already meet or exceed their dynamic thresholds on their own, the augmentation phase is skipped entirely.

6. **Active Learning and Manual Labeling:** To overcome the bottleneck of massive manual annotation and verify the quality of the dataset, the pipeline deploys the EASE framework [6]. Instead of randomly labeling data, the system uses informative and diverse sampling to selectively flag highly uncertain and non-redundant textual edge cases. These filtered samples are then routed to human annotators who perform manual labeling, acting as the "oracle" to provide definitive ground-truth sentiment labels for the most ambiguous code-switched text while drastically reducing overall human effort.
7. **Multilingual Transformer Fine-Tuning:** A pre-trained Cross-lingual Language Model (XLM-RoBERTa) serves as the core linguistic engine. In this phase, the model is actively fine-tuned using the study's constructed dataset to accurately interpret the contextual relationships and complex syntax of the localized, code-switched text. Building on the framework established by Maceda et al. [32], this study introduces several architectural modifications. Where the baseline study used a static, manually annotated dataset for general ternary sentiment classification, this pipeline works with a dynamic, hybrid dataset that combines human-annotated edge cases and strictly regulated LLM-generated synthetic data.
8. **Classification (Aspect-Level Sentiment Prediction):** Once the target aspects have been extracted and the model fine-tuned, the pipeline evaluates the emotional polarity attached to each individual feature [47]. The model classifies the sentiment specific to that isolated review-aspect pair, enabling a highly nuanced and granular understanding of consumer feedback [46][7].

3.3 Output Phase

The output phase represents the final deliverables after the process. This will present Aspect-level polarities, Consumer insights, and Local Market Intelligence, transforming raw, unstructured feedback into metrics. Relying on an ambiguous average star rating often fails to provide specific operational insights. Extraction of granular aspect-level polarities allows consumers to compare establishments [25]. These extracted insights provide businesses with market intelligence enabling them to pinpoint operational strengths and address targeted areas for improvement [34] [46].

The conceptual framework demonstrated how challenges of processing large volumes of unstructured, code-switched local review can be effectively managed. Routing raw text through a robust and detailed pipeline of active learning, augmentation, and

transformer-based aspect classification. It bridges the gap between online reviews and structured, actionable intelligence. Further discussions of technical configurations, procedures, and executions will be detailed in the succeeding methodology.

3.4 Methodology

3.4.1 Research Design

The research follows a multi-stage computational pipeline designed to perform Aspect-Based Sentiment Analysis (ABSA) on code-switched consumer feedback. The architecture integrates transformer-based deep learning, generative artificial intelligence for taxonomy discovery, and human-in-the-loop active learning to provide granular market intelligence.

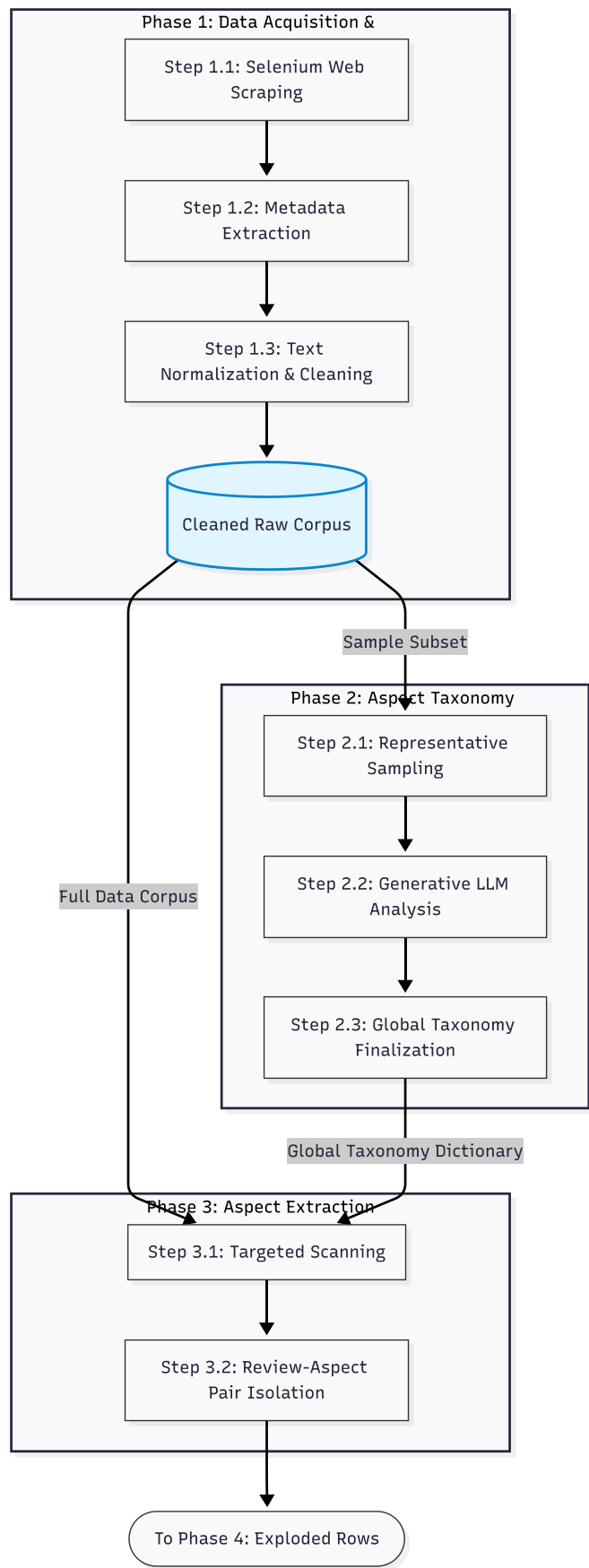


Figure 10: *Data Acquisition and Aspect Extraction*

Phase 1: Data Acquisition and Textual Preprocessing

The initial phase, as shown in Figure 10, is the automated harvesting and textual normalization of the primary research corpus.

- **Step 1.1: Automated Web Harvesting:** To bypass the limitations inherent in standard API rate-limiting, a custom scraping utility utilizing the *Selenium WebDriver* is deployed. This allows for the continuous retrieval of review histories from Google Maps for target coffee and matcha establishments in Davao City.
- **Step 1.2: Metadata Extraction:** For each retrieved entry, the system extracts the raw review text alongside the metadata, including numerical star ratings, temporal markers (timestamps), and establishment identifiers.
- **Step 1.3: Textual Normalization:** Utilizing the *Pandas* library and Regular Expressions (Regex), the corpus undergoes rigorous cleaning. This process involves the removal of HTML artifacts, normalization of idiosyncratic characters, and the conversion of emoticons into descriptive text to preserve semantic sentiment cues during model training.

Phase 2: Aspect Taxonomy

Rather than adopting a static industry standard taxonomy or relying on traditional unsupervised topic modeling like Latent Dirichlet Allocation (LDA), which already tends to struggle with the noisy, informal syntax of code-switched regional text, this research uses a direct generative approach to identify localized operational dimensions instead. Recent work by Boughanmi et al. [9] shows that large language models can directly produce semantic taxonomies that match or even outperform human coding, without needing an intermediate LDA step.

- **Step 2.1: Representative Stratified Sampling:** A subset of normalized reviews is randomly sampled to serve as a representative seed for exploratory analysis.
- **Step 2.2: Generative Semantic Analysis:** The sample is processed via a Generative Large Language Model (Gemini 3). The prompt design explicitly instructs the model to act as a qualitative researcher, tasked with identifying recurring linguistic patterns, frequent noun-phrases, and implicit themes specific to the regional Davao City consumer context (e.g., isolating Bislish terms for taste or ambiance).
- **Step 2.3: Global Taxonomy Formalization and Validation:** A master dictionary of aspects is generated based on the LLM output. Before deployment, this dictionary undergoes a human-in-the-loop validation check by the researchers to ensure context-specific dimensions, such as [Parking Availability] and [Wi-Fi Reliability], are accurately captured alongside traditional metrics.

Phase 3: Granular Aspect Extraction and Data Decomposition

To achieve high-resolution sentiment mapping, unstructured feedback is decomposed into granular units. Refer to Figure 10.

- **Step 3.1: Targeted Scanning:** An extraction algorithm utilizes the formal taxonomy to scan the comprehensive review dataset for explicit and implicit aspect mentions.
- **Step 3.2: Review-Aspect Pair Isolation:** This process, referred to as the "Data Explosion," partitions multi-aspect reviews into structured rows. Each row maps a specific textual segment to a single target aspect (e.g., $[Text] + [TargetAspect]$). This prevents "sentiment cancellation," in which a single review shows conflicting polarities across different operational dimensions.

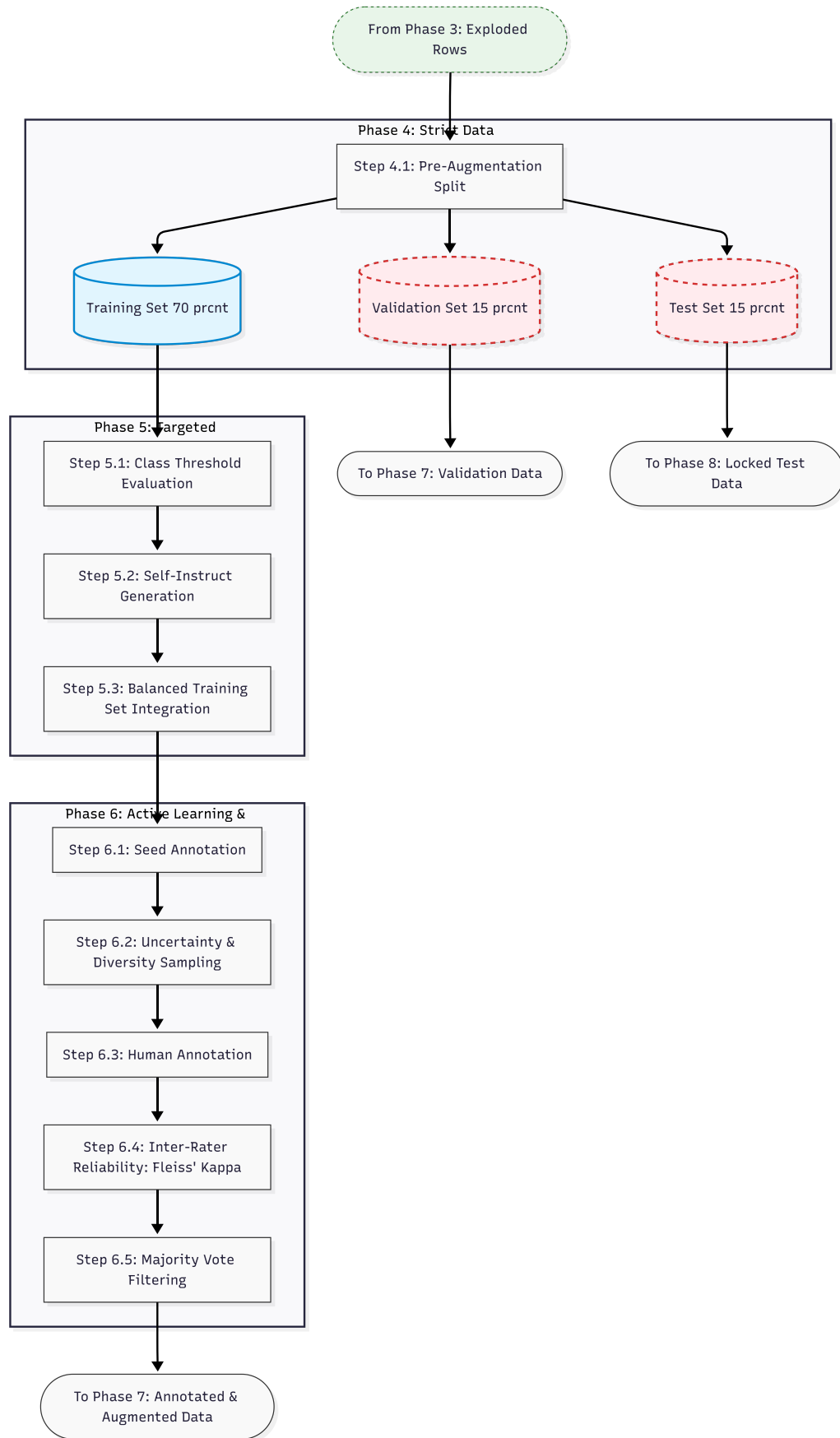


Figure 11: Partitioning, Augmentation and Annotation

Phase 4: Strict Data Partitioning

To maintain experimental integrity and preclude data leakage, strict partitioning is enforced, as shown in Figure 11.

- **Step 4.1: Pre-Augmentation Partitioning:** The authentic dataset is partitioned into Training (70%), Validation (15%), and Test (15%) sets. This was done to prevent data leakage and maintain experimental integrity.
- **Step 4.2: Evaluation Firewall:** The Validation and Test sets are isolated from all synthetic data and training iterations. This ensures that final performance metrics represent the model’s genuine generalizability to authentic human language.

Phase 5: Targeted Synthetic Data Augmentation

To mitigate class imbalance found in niche operational categories, synthetic augmentation is utilized strictly as a fallback mechanism, shown in Figure 11.

- **Step 5.1: Empirically Determined Threshold (Learning Curve):** Rather than setting a fixed, arbitrary threshold for data sufficiency, the minimum number of instances needed for each aspect sentiment pair will be determined empirically. A small sample of authentic data will be trained incrementally in batches (50, 100, and 150 instances) [44], and the resulting F1 scores will be plotted to produce a learning curve. The point where adding more data stops meaningfully improving performance, where the curve levels off, is what will define the minimum viable threshold for that specific class [23][41].
- **Step 5.2: Self-Instruct Generation:** If a specific category remains underrepresented and cannot naturally meet the empirically determined threshold, a Large Language Model (Gemini 3) will be utilized. It will use the Self-Instruct framework to synthesize additional training samples. Authentic reviews serve as “seed context” to ensure that the synthetic output accurately replicates the regional Bislish and Taglish syntactic nuances.
- **Step 5.3: Training Set Integration:** Synthetic data is merged exclusively into the Training set, resulting in a balanced corpus that prevents model bias toward high-frequency categories while protecting the integrity of the validation sets.

Phase 6: Active Learning with Human Annotation

Annotation efficiency is maximized through an iterative labeling process (see Figure 11).

- **Step 6.1: Initial Seed Annotation:** Domain-Competent Peer Annotators, comprising undergraduate students from Ateneo de Davao University who are native speakers of the regional Bislish and Taglish dialects, will perform manual sentiment labeling on a small, random initial batch of the exploded Training data. Their localized linguistic fluency qualifies them to accurately interpret the highly contextual code-switching present in the corpus. This will establish the baseline ground truth required to initialize the system.

- **Step 6.2: Uncertainty and Diversity Sampling:** A preliminary baseline instantiation of the XLM-RoBERTa model is trained on the seed data to act as the active learning querying engine. This in-loop model flags "low-confidence" (predictions below the 75% threshold) and non-redundant samples [37]. These instances are prioritized for human intervention, ensuring that manual labeling efforts are focused strictly on the most ambiguous segments rather than redundant data [6].
- **Step 6.3: Human Labeling:** The flagged edge cases are routed back to the Domain-Competent Peer Annotators for a targeted round of manual annotation.
- **Step 6.4: Inter-Rater Reliability (Fleiss' Kappa):** The overall consistency of the annotations is mathematically evaluated using the Fleiss' Kappa statistic. To ensure high-quality ground truth, the study aims for a rigorous standard. If the batch achieves a score of $k \geq 0.61$ (Substantial Agreement) [29], it is validated and proceeds to the next step. If a batch falls below this threshold, it triggers a mandatory recalibration phase where annotators must review discordant instances, refine guidelines, and re-annotate until the substantial reliability threshold is met.
- **Step 6.5: Majority Vote Filtering:** For the mathematically validated batch, individual ground-truth labels are finalized using a strict majority-voting mechanism. Instances of absolute disagreement are considered highly ambiguous and permanently discarded to prevent noise from being introduced during the fine-tuning phase.

Since active learning runs as an iterative cycle, a stopping criterion is enforced to keep the process from querying indefinitely. The annotation loop repeats until the model reaches Confidence Stability, which is the point where the remaining unlabeled pool no longer produces instances that fall below the uncertainty threshold. As a logistical failsafe, the loop will also stop if it hits a predefined maximum annotation budget, essentially a hard cap on the total number of instances based on the time and availability of the volunteer peer annotators. Once either condition is met, the active learning phase ends and the dataset moves to Phase 7.

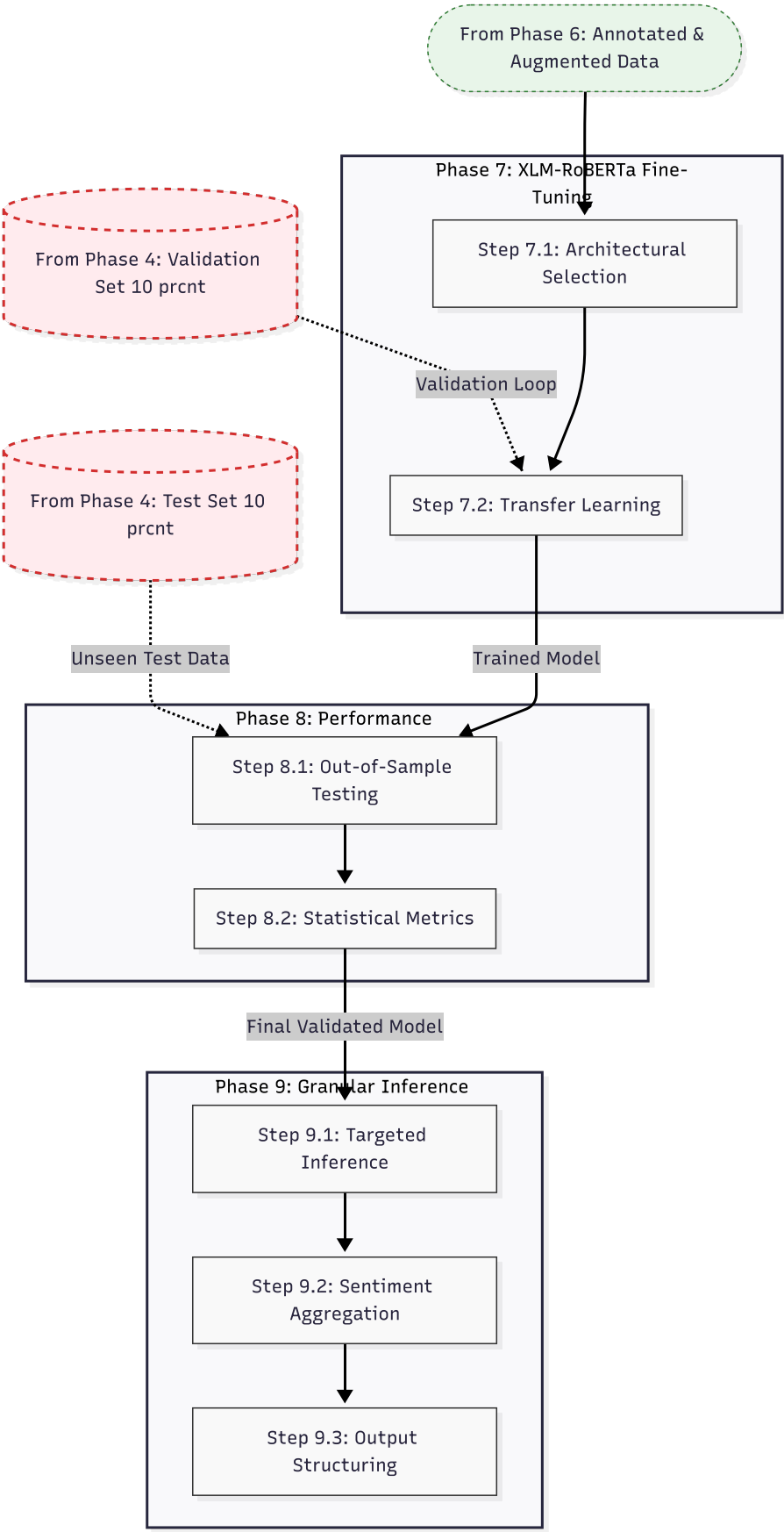


Figure 12: Modeling and Evaluation

Phase 7: XLM-RoBERTa Fine-Tuning

The core sentiment engine is optimized for the code-switched linguistic landscape of the Philippines (see Figure 12).

- **Step 7.1: Architectural Selection:** The research utilizes the *XLM-RoBERTa-base* architecture via the *Hugging Face Transformers* library. XLM-RoBERTa is selected over mBERT due to its superior performance on informal web-scraped text and its extensive pre-training on a 2.5TB corpus that includes Cebuano and Tagalog [45].
- **Step 7.2: Transfer Learning:** The model undergoes fine-tuning training set, learning to associate complex code-switched syntax with specific aspect-level sentiment polarities.

Phase 8: Performance Evaluation and Validation

The pipeline’s effectiveness is validated against the actual data (see Figure 12).

- **Step 8.1: Out-of-Sample Testing:** The fine-tuned XLM-RoBERTa model is evaluated using the locked Test set, which contains 100% authentic, unseen human feedback.
- **Step 8.2: Statistical Metrics:** Performance is quantified through F1-Score, Precision, and Recall. Success is defined by the model’s ability to accurately classify sentiment across varied aspects without overfitting on synthetic training data.

Phase 9: Granular Inference and Insight Aggregation

The final phase translates the computational outputs into structured, actionable insights designed to support decision-making for both consumers and business owners (see Figure 12).

- **Step 9.1: Targeted Inference:** The validated model processes authentic feedback for individual target establishments to generate aspect-level sentiment predictions.
- **Step 9.2: Sentiment Aggregation:** The predicted polarities are compiled to provide a comprehensive, quantitative overview of consumer satisfaction. The system aggregates the counts of positive, negative, and neutral sentiments for each discovered operational dimension and calculates their prevailing sentiment percentages (e.g., determining that 85% of reviews mentioning “Wi-Fi Reliability” are positive).
- **Step 9.3: Output Structuring (Granular System Interface):** Rather than returning raw computational output, the final vision for system integration is to present these aggregated metrics in a structured, consumer facing format. The system will display a directory style listing of local coffee and matcha establishments, replacing vague average star ratings (4.2 stars, 4.0 stars) with granular, aspect-specific scores. A specific shop might show a 75% positive sentiment for Taste but a 60% negative sentiment for Parking Availability. Following the principle of

translating model outputs into something practically useful and easy to read, this structured output works as a proof of concept.

3.4.2 Data Collection

The primary dataset for this study consists of publicly available consumer reviews of coffee and matcha establishments in Davao City, collected from Google Maps. Although the Google Places API offers a structured way to retrieve this kind of data, it limits results to roughly five reviews per establishment. Chen et al. (2024) [12] and Trezza (2023) [49] both noted that this restriction makes APIs impractical for training machine learning models, and this study ran into the same problem. A custom web scraper will be built to collect enough review data to fine-tune the XLM-roBERTa model (Figure 13).

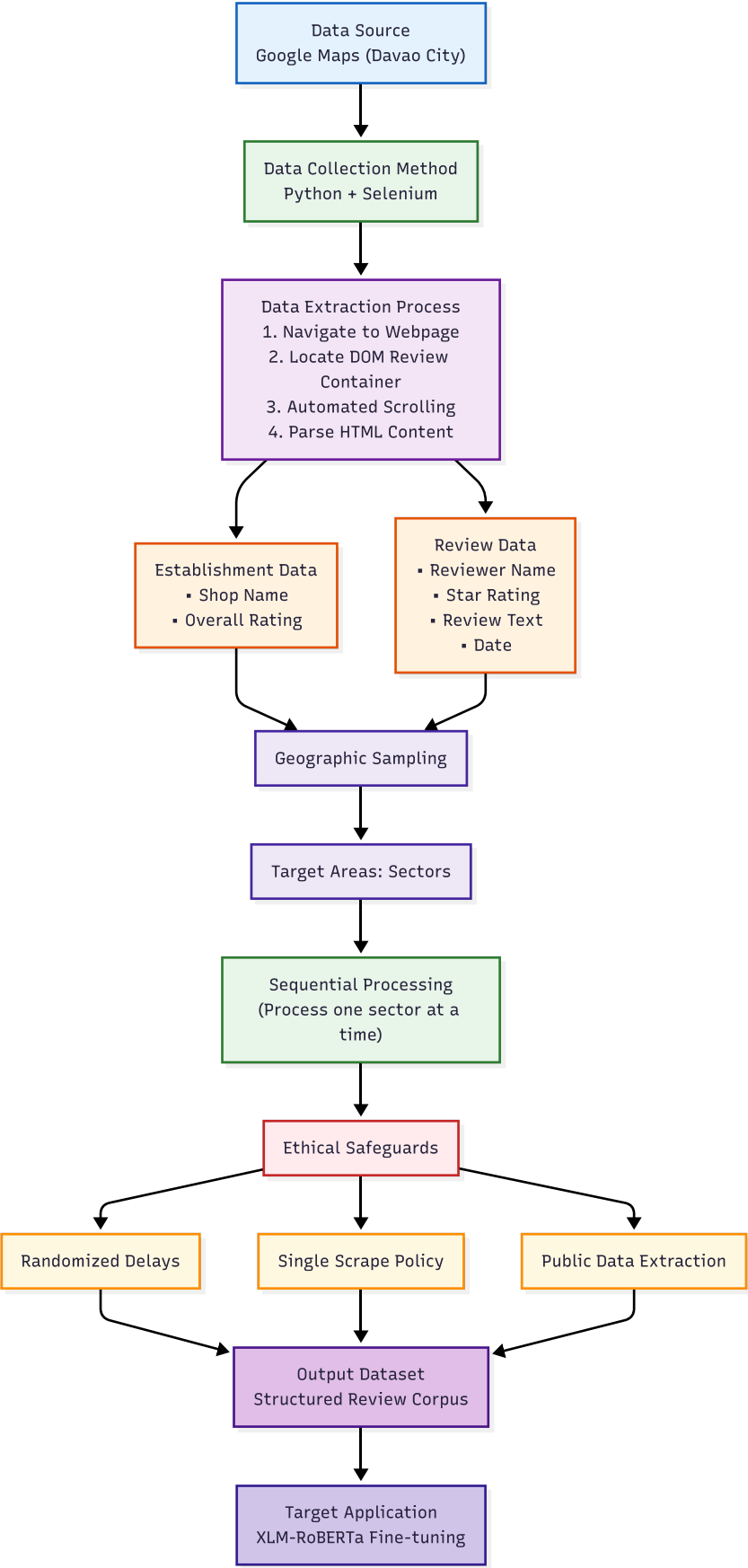


Figure 13: Data Collection Process

Extraction Architecture and Logic

Data collection will be carried out using a Python script built on Selenium WebDriver. Selenium will be chosen since Google Maps loads its content dynamically through JavaScript, which means static parsers like BeautifulSoup would return nothing useful. The extraction will follow a straightforward sequence. The WebDriver will first navigate to the Google Maps page of each target establishment. It will then locate the DOM container where reviews are stored. Because Google Maps only loads older reviews as the user scrolls down, the script will run an automated loop that keeps scrolling the review pane until no new content appears. Once all reviews are loaded, the script will parse the HTML to extract the target shop's name and overall rating, alongside granular data for each entry, including the reviewer's name, individual star rating, full review text,, and the date it will be posted.

Systematic Geographic Sampling

To cover Davao City without making the data collection process unmanageable, the study will divide the city into key commercial sectors such as Ecoland, Matina Crossing, Poblacion District, and others. The script will target all identified coffee and matcha shops within one sector at a time before moving on to the next, which will keep the process organized and reduce the risk of gaps in coverage.

Ethical Considerations and Technical Safeguards

Automated scraping can put unintended load on servers and trigger IP blocks if done carelessly, so the data collection will follow what is commonly called polite scraping. The script will insert randomized time delays between scroll actions and page requests so the behavior will more closely resemble a human browsing the site. Each establishment will also be scraped only once rather than on a recurring schedule (every 3 hours) , which will avoid putting repeated strain on Google's servers. All of the data collected is already publicly visible on Google Maps and will be used only for this academic study with no commercial intent.

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